AEM313 Airfoil Project

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A study of NACA 4-digit airfoils was conducted using thin airfoil theory (TAT) and XFOIL. A total of six airfoils, consisting of two assigned baseline airfoils and four additional selected airfoils, were studied to learn the effects of maximum camber and camber location. The angle of zero lift, moment coefficient about the leading edge, and center of pressure of the airfoils were calculated using TAT. Inviscid and viscous results were obtained using XFOIL at a Reynolds number of 10^6. The selected airfoils were studied further for Reynolds number and thickness variations in them. The trends of the lift-curve behavior and pressure-coefficient behavior were compared to each other to evaluate how airfoil geometry and flow conditions influence them. These results were used to identify how camber, camber location, Reynolds number, and thickness of the airfoils affect their aerodynamic performance.

I. Introduction

This airfoil study carefully examines the aerodynamic behavior of four selected NACA 4-digit airfoils and two assigned baseline NACA 4-digit airfoils using TAT and XFOIL. Group 32 was assigned NACA 1108 and NACA 2408 as our baseline airfoils, and NACA 4408, NACA 0008, NACA 1408, and NACA 2308 were selected as additional airfoils. The additional airfoils were used to study the effect of maximum camber and camber location. The thickness was kept fixed for the TAT portion of the additional airfoil analysis. The main question that is being addressed in this study is how changes in the geometry of an airfoil can affect the angle of zero lift, pitching-moment characteristics, center of pressure, and lift-curve behavior. TAT results and XFOIL inviscid and viscous solutions at a Reynolds number of 10^6 are compared, and the additional selected airfoil cases are examined further to learn about the effects of Reynolds number and thickness variation.

II. Procedure

TAT and XFOIL were used to perform this airfoil study of the set of six NACA 4-digit airfoils. Group 32 was assigned the following baseline airfoils: NACA 1108 and NACA 2408. Four additional airfoils were selected to learn about the effects of maximum camber and camber location. This was done while avoiding any changes in thickness in the TAT portion of the airfoil analysis. Also, the mean camber line was defined using the standard NACA 4-digit camber-line equation for the additional selected airfoils. The angle of zero lift, moment coefficient about the quarter-chord, moment coefficient about the leading edge, and center of pressure were determined using TAT from this relation. Handwritten TAT calculations were done by each member of the group and are shown in the appendix.

TAT assumes incompressible, inviscid, irrotational flow over a thin airfoil with the following attributes: a small camber and a small angle of attack. TAT was used to study the changes in camber and camber location because of these TAT assumptions; however, it was not used to study the thickness effects. The results obtained from the TAT analysis were then compared to the numerical results from XFOIL.

XFOIL is a panel-method airfoil analysis program used in this airfoil study to generate both inviscid and viscous aerodynamic predictions. For each of the six airfoils, lift-curve data were obtained using XFOIL for both inviscid and viscous flow at a Reynolds number of 10^6 on the default program settings. The resulting data generated by XFOIL was then exported to EXCEL and plotted to compare the C_L versus angle of attack for TAT and XFOIL. Pressure-coefficient distributions were examined as well for the selected airfoil cases to compare surface pressure behavior under inviscid conditions versus viscous conditions.

Additional XFOIL studies were conducted on two selected airfoil shapes from the original set of six airfoils. For these two selected airfoils, viscous-flow effects were evaluated by varying Reynolds number above and below 10^6, and thickness effects were studied by varying thickness with a fixed camber shape to separate the thickness effects from the camber effects. The following input values were recorded for each case and used consistently throughout the analysis: Reynolds number, angle-of-attack range, and airfoil designation.

In Xfoil, the airfoils were analyzed by defining both geometric and operating parameters prior to the solution. The geometry was prepared using the GDES command when needed to alter the shape of the airfoil. Flow conditions were set and angles of attack were specified and studied using the OPER menu, where the viscous and inviscid cases were studied. Thickness variation was done using the TSET command, that allows the user to change the thickness and camber of the base airfoil. Aerodynamic performance was analyzed over a set range of angles of attack using the ASEQ command, angles of attack used were ranged from -5 to 20 degrees in increments of .5, while

individual cases were also analyzed using the ALFA command for specific angles of attack. These inputs allowed the generation of the lift, drag, and pressure coefficient data for both our inviscid and viscous cases across all of the airfoils studied.

The principal equations that were used in the analytical portion of the airfoil study were the NACA 4-digit mean camber-line equation and the TAT relations used to determine the following: angle of zero lift, quarter-chord moment coefficient, leading-edge moment coefficient, and center of pressure. The default panel settings were used in XFOIL, and each airfoil was analyzed for inviscid flow and for viscous flow at a Reynolds number of 10^6. Lift-curve data was then generated over the selected angle-of-attack range to be compared with the TAT results.

The NACA 4-digit mean camber line equation for $x/c \leq p$ is $z/c=(m/p^2)[2p*x/c – (x/c)^2]$ and for $x/c \geq p$ it is $z/c=(m/(1-p)^2)[(1-2p)+2p*x/c – (x/c)^2]$. The camber line slope was obtained using dz/dx. The thin airfoil transformation equation is $x/c=(1-\cos\theta)/2$. The Fourier coefficient equations are $A_0= \alpha – (1/\pi) \int_0^\pi (dz/dx)\, d\theta$ and $A_n=(2/\pi) \int_0^\pi [(dz/dx)*\cos(n\theta)d\theta$. The lift coefficient equation is $C_L=2\pi(\alpha- \alpha_{L=0})$. The angle of zero-lift equation is $\alpha_{L=0}=(1/\pi) \int_0^\pi [(dz/dx)(1-\cos\theta)]d\theta$. The quarter-chord moment coefficient equation is $C_{m,c/4}=(-\pi/4)(A_1-A_2)$. The leading-edge moment coefficient equation is $C_{m,LE}= C_{m,c/4} – 1/4(C_L)$. The center of pressure equation is $x_{cp}/c= -(C_{m,LE}/ C_L)$.

The work for this airfoil study was divided between the two members of Group 32 so that both students completed an XFOIL analysis and evenly contributed to the airfoil calculations and project report. Collin Atkins contributed the following: an XFOIL analysis, calculations and plots for NACA 2408 (assigned baseline airfoil), calculations and plots for NACA 1408 (selected additional airfoil), calculations and plots for NACA 2308 (selected additional airfoil), abstract, introduction, procedure, Cp plots, NACA 2408 thickness variation plots, lift-curve plots, MCL plot, Tables 1 and 2, formatting for report, handwritten TAT calculation, and EXCEL figures. Nigel Pruitt completed an XFOIL analysis, worked on NACA 1108 (assigned baseline airfoil), NACA 4408 (selected additional airfoil), and NACA 0008 (selected additional airfoil), and wrote the discussion and conclusion for the project report. Both students also contributed to the results and appendix sections of the project report.

III. Results

A. Mean Camber Lines

The mean camber lines for NACA 2408, 1408, 2308, 1108, 4408, and 0008 are shown in Fig. 1. In the plot, as expected, NACA 1408 has a small camber magnitude compared to NACA 2408. And so, its camber line remains below that of NACA 2408 over most of the chord. NACA 2308 has the same maximum camber as NACA 2408; however, its maximum camber occurs farther forward, which shifts the peak of the camber line towards the leading edge of the airfoil. A different trend was observed for NACA 4408 and NACA 1108. NACA 1108's maximum camber occurs farther along the chord. NACA 4408 has the same camber location as NACA 2408; however, NACA 4408 has a larger magnitude. On the other hand, NACA 0008 is symmetric; therefore, it has zero camber over the entire chord.

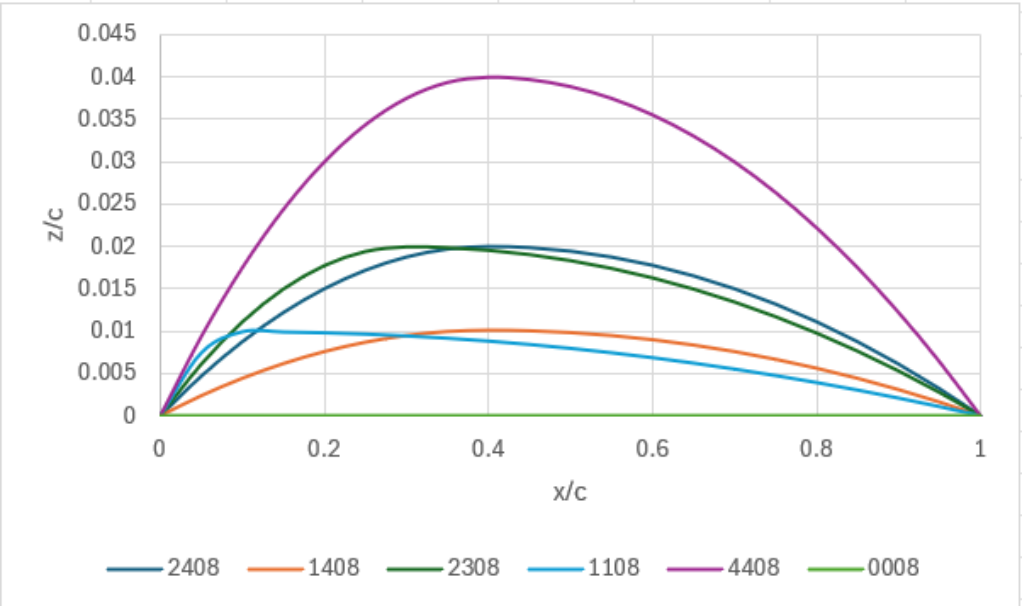


Fig. 1. Mean Camber Lines for NACA 2408, 1408, 2308, 1108, 4408, and 0008

B. Airfoil Geometry

A summarization of the six NACA 4-digit airfoils used in the airfoil study is shown in Table 1. Group 32 was assigned NACA 1108 and NACA 2408 as baseline airfoils. The other four additional airfoils were selected by Group 32 themselves to compare the effects of maximum camber and camber location while maintaining a constant thickness ratio of 0.08.

Table 1. Geometric Parameters and Comparison Purpose for NACA 1108, 2408, 1408, 2308, 4408, and 0008

Airfoil	Maximum Camber, m	Camber Location, p	Thickness, t/c	Comparison
NACA 1108	**0.01**	**0.10**	**0.08**	**Assigned baseline airfoil; compares very forward camber location**
NACA 2408	**0.02**	**0.40**	**0.08**	**Assigned baseline airfoil**
NACA 1408	**0.01**	**0.40**	**0.08**	**Compares the effect of reduced camber relative to NACA 2408**
NACA 2308	**0.02**	**0.30**	**0.08**	**Compares the effect of a more forward camber location relative to NACA 2408**
NACA 4408	**0.04**	**0.40**	**0.08**	**Compares the effect of increased camber relative to NACA 2408**
NACA 0008	**0.00**	**0.00**	**0.08**	**Symmetric reference airfoil for zero-camber comparison**

C. TAT Results

The TAT results for the six NACA 4-digit airfoils that were analyzed in the airfoil study are listed in Table 2. Table 2 shows how changes in camber magnitude and camber location in an airfoil affect the angle of zero lift, pitching moment, and center of pressure. These TAT result values were then used as a baseline in the comparison with the XFOIL theoretical predictions.

Table 2. TAT results for the six NACA airfoils used in this study at an angle of attack of zero degrees

Airfoil	$\alpha_{L=0}$	$C_{m,c/4}$	$C_{m,LE}$	X_{cp}/c
1108	**-0.8575**	**-0.0145**	**-0.0380**	**0.4043**
2408	**-2.0772**	**-0.0531**	**-0.1101**	**0.4832**
1408	**-1.0386**	**-0.0266**	**-0.0550**	**0.4832**
2308	**-1.9180**	**-0.0447**	**-0.0973**	**0.4626**
4408	**-4.1545**	**-0.1064**	**-0.3200**	**0.4836**
0008	**0.0000**	**0.0000**	**0.0000**	**undefined**

D. Lift-Curve Comparisons

Figures 2-7 show the lift coefficient (CL) as a function of the angle of attack for the NACA 2408, 1408, 2308, 1108, 4408, and 0008 airfoils, comparing the thin airfoil theory (TAT) calculations done by hand with the XFOIL panel method results. The lift curve exhibits an expected linear relationship with the angle of attack. The XFOIL correctly predicted the approximate angle of zero lift ($\alpha_{L=0}$) for all airfoils. The panel method also nearly perfectly predicted the lift slope for each airfoil, with the slope from XFOIL prediction being just slightly higher than that of the TAT hand calculations across all airfoils studied. Because XFOIL captures more detailed geometry details, there are small differences in predicted lift compared to that of TAT.

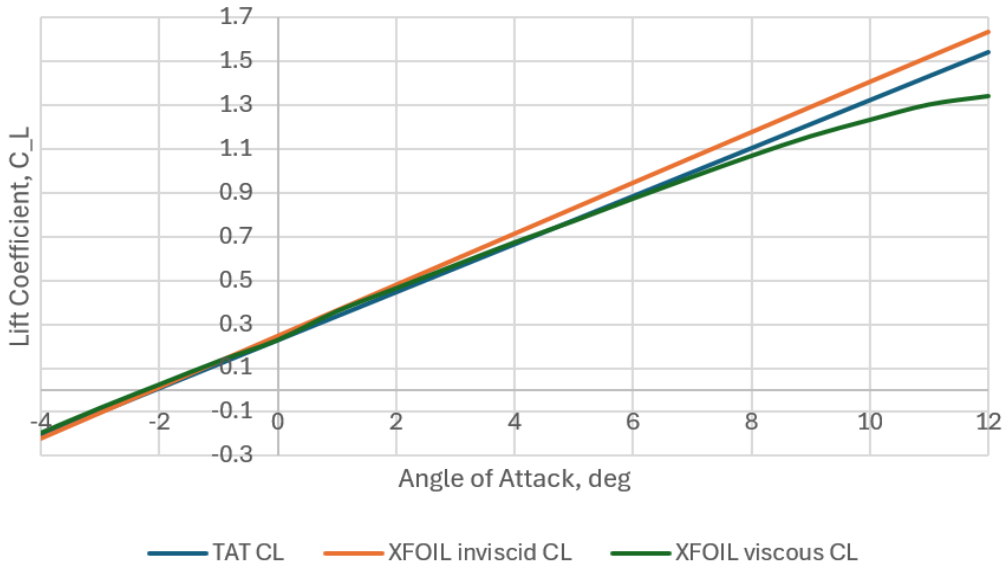


Fig. 2. Lift-Curve Comparison for NACA 2408 using TAT, XFOIL Inviscid, and XFOIL Viscous Results

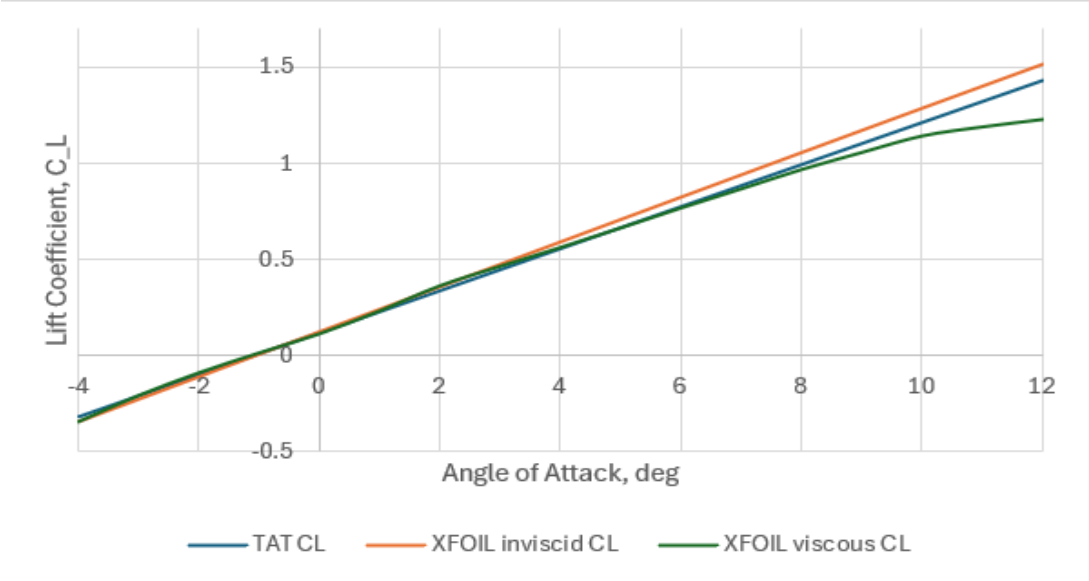


Fig. 3. Lift-Curve Comparison for NACA 1408 using TAT, XFOIL Inviscid, and XFOIL Viscous Results

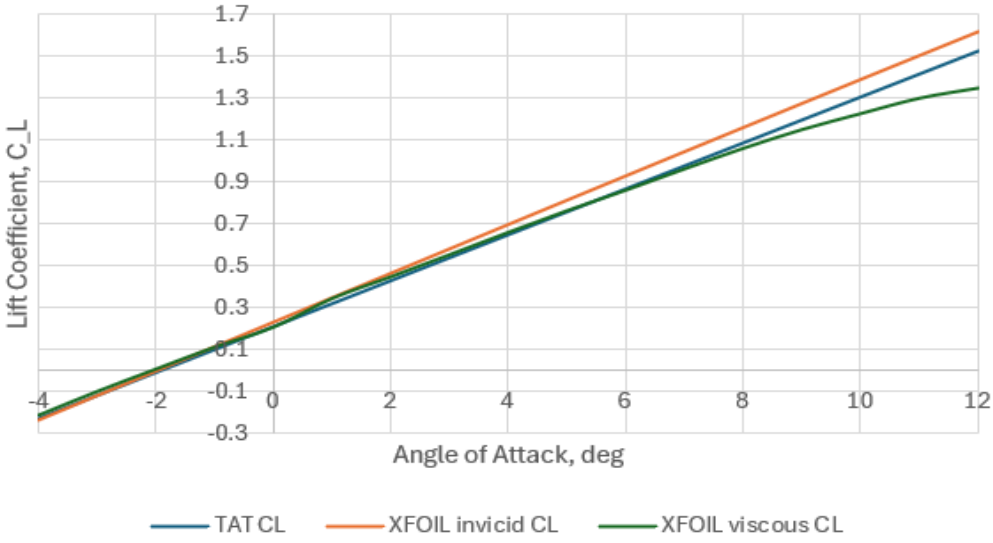
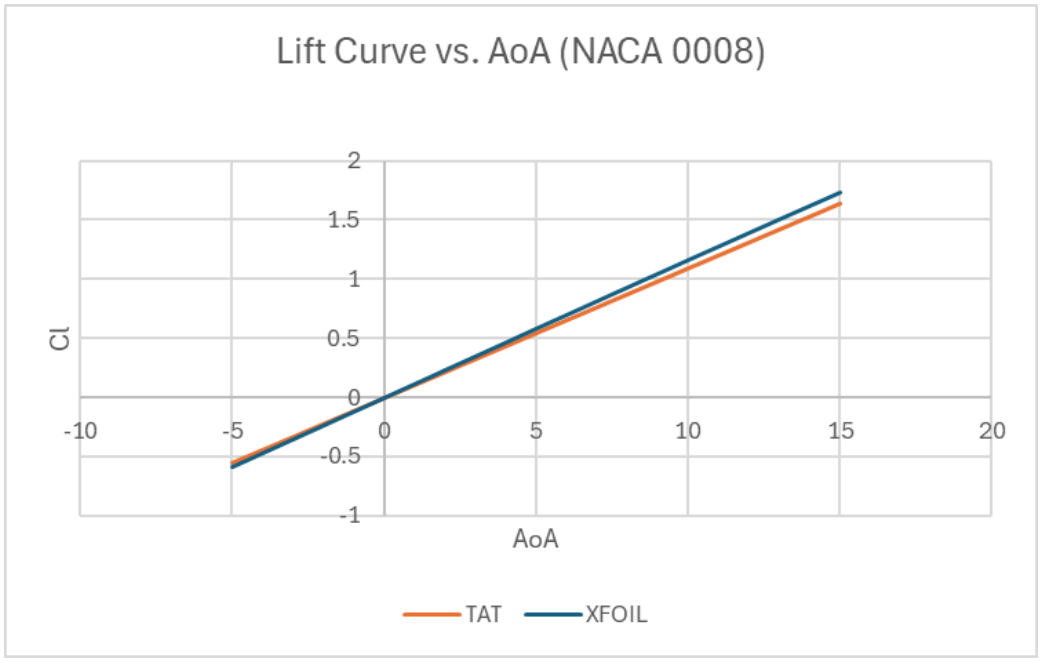


Fig. 4. Lift-Curve Comparison for NACA 2308 using TAT, XFOIL Inviscid, and XFOIL Viscous Results



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Fig. 5. Xfoil vs TAT Lift curve (NACA 0008)

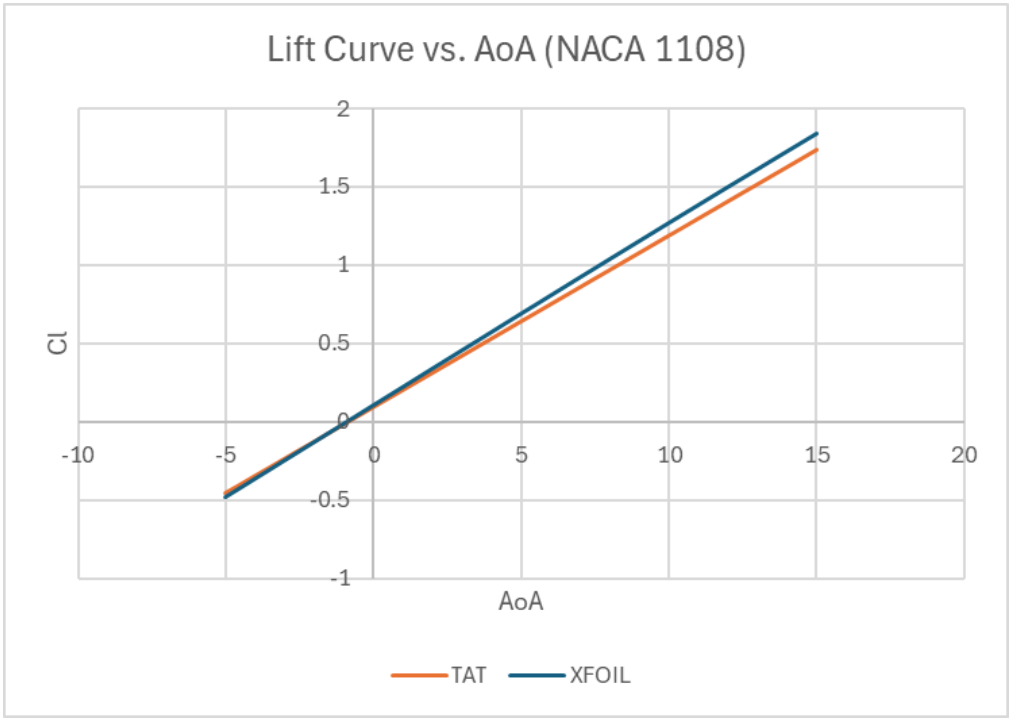


Fig. 6. Xfoil vs TAT Lift curve (NACA 1108)

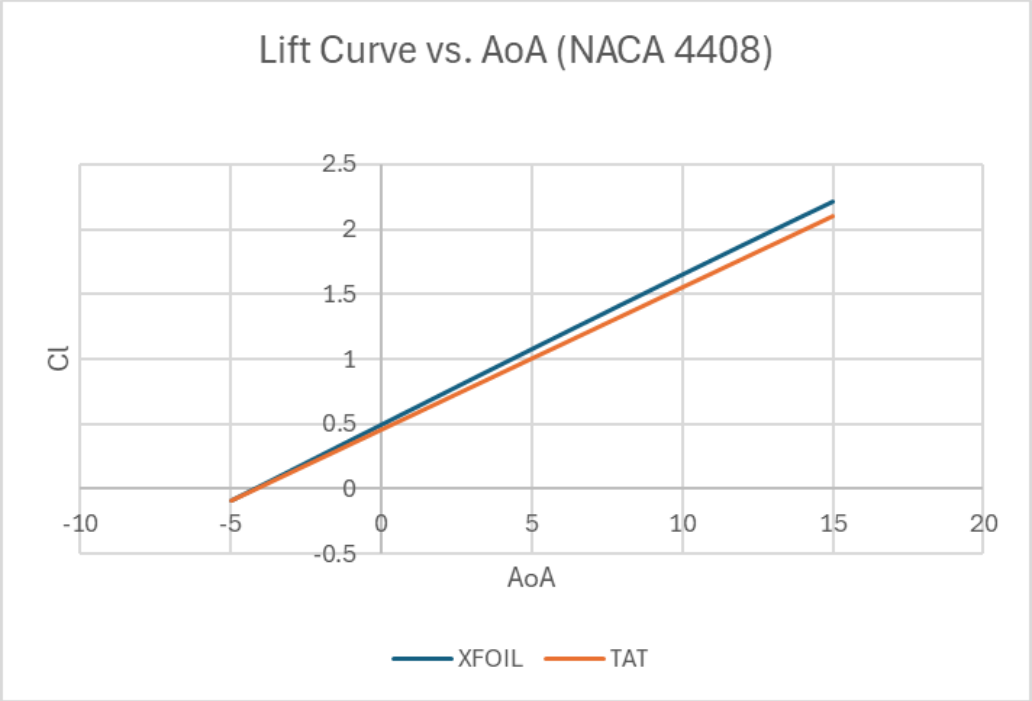


Fig. 7. Xfoil vs TAT Lift curve (NACA 4408)

Figures 2-4 and 8-10 show how the lift coefficient (Cl) as a function of angle of attack for the selected airfoils is affected by the introduction of viscosity by comparing inviscid and viscous XFOIL results at a Reynolds number of 10e^6. For all airfoils, both cases exhibit an identical linear increase in the lower angles of attack. At around an angle of attack of 4 degrees, is where the viscous case starts to diverge, showing a sort of parabolic curve, with the peak of which is the stall point of the airfoil. Overall, the viscous results provide a more realistic view of airfoil performance, specifically at higher angles of attack, where flow separation is more important

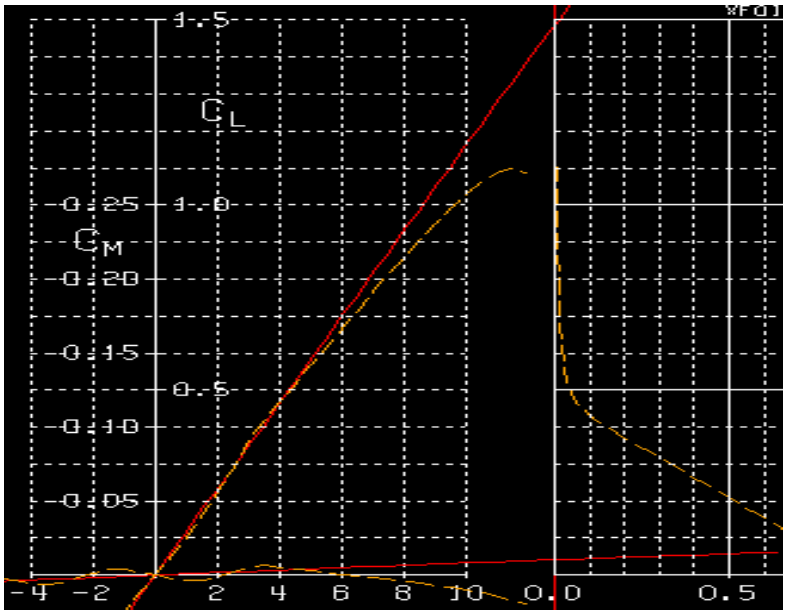


Fig. 8. Inviscid vs. Re=10e^6 Lift curve NACA 0008

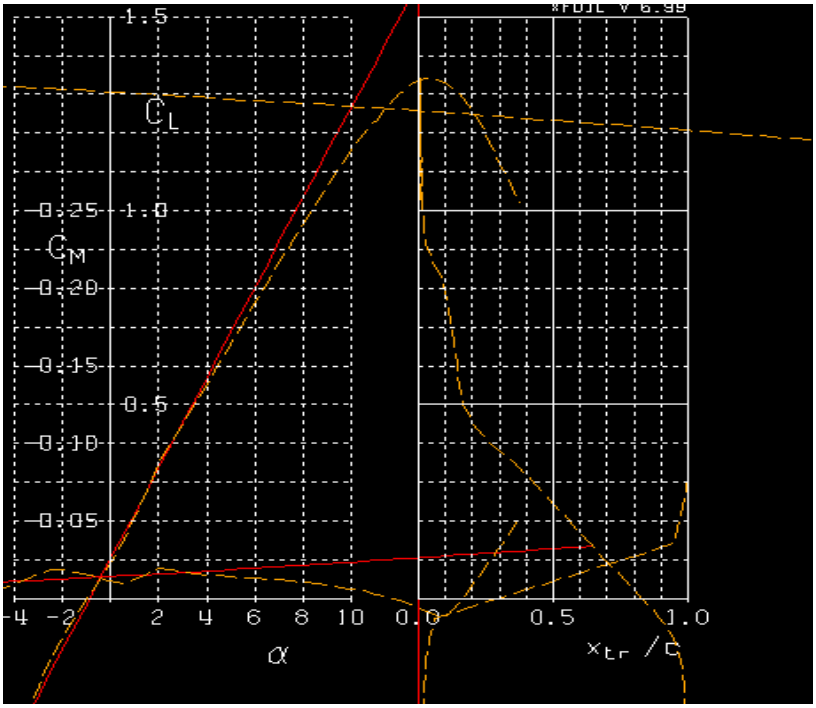


Fig. 9. Inviscid vs. Re=10e^6 Lift curve NACA 1108

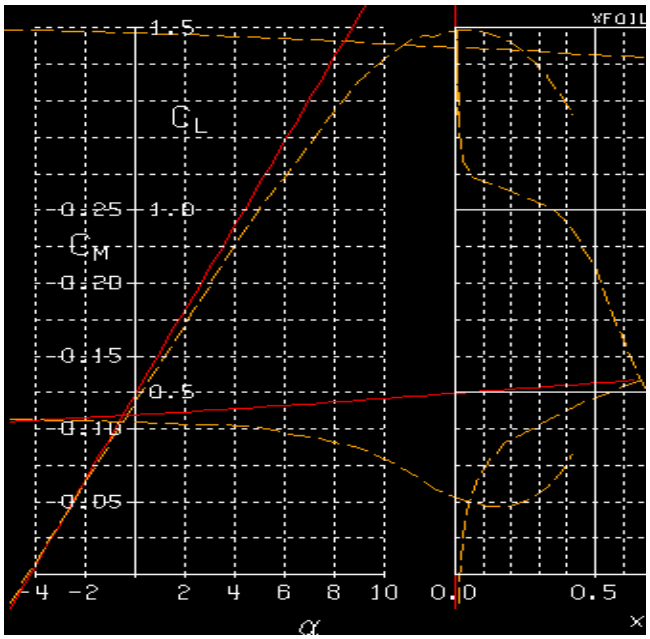


Fig. 10. Inviscid vs. Re=10e^6 Lift curve NACA 4408

E. Re Variation

Figure 11 shows the lift coefficient (Cl) as a function of the angle of attack for the NACA 1108 at Reynolds numbers of 10e4, 10e5, and 10e6. All the cases showed a similar initial linear increase in lift in the lower angles of attack. The clear differences start to be observable at higher angles. At the lowest Reynolds number (10e4), the lift curve has the smallest slope and begins to peak at the airfoil's stall point earliest. At the highest Reynolds number of 10e7, the airfoil produces the greatest amount of lift and maintains a linear behavior to higher angles of attack than the rest. This trend occurs because increasing the Reynolds number also reduces the relative thickness of the boundary layer, which delays flow separation, allowing the airfoil to be able to attain higher lift.

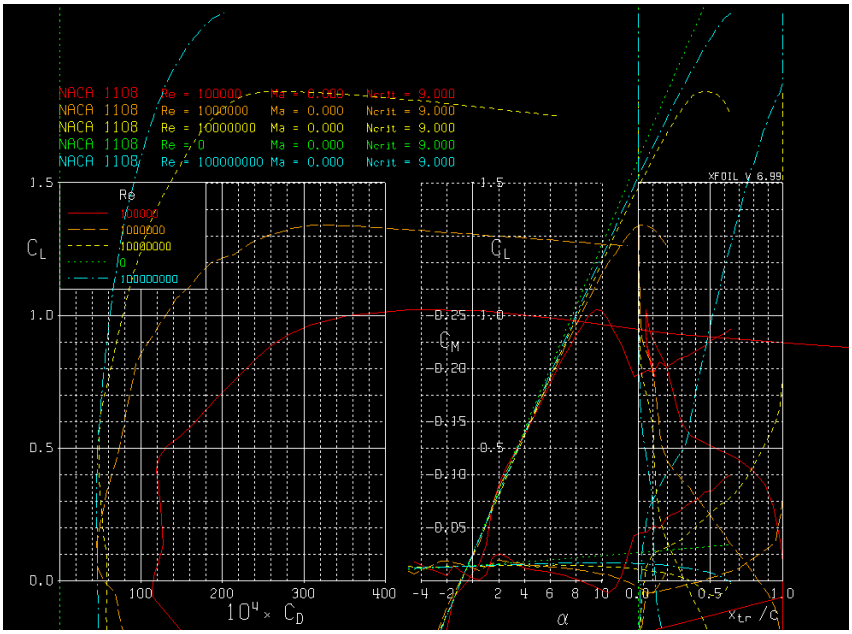


Fig. 11. Re variation (NACA 1108)

F. Thickness Variation

Figures 12-14 show the coefficient of lift (Cl) as a function of the angle of attack for the NACA 11xx and NACA 24xx cambered airfoils. All the cases showed a similar initial linear increase in the lower angles of attack. The differences begin to be observable as the thickness of the airfoils increases or decreases. Thicker airfoils generally produce slightly higher lift coefficients at a given angle of attack. At higher angles of attach, the thicker airfoils delay stall and exhibit a more gradual lift drop off, while thinner airfoils experience sharper stall behavior. This trend most likely is because increasing an airfoil's thickness improves the airfoil's ability to sustain pressure difference over the surface, which helps slow down flow separation and allow for higher lift.

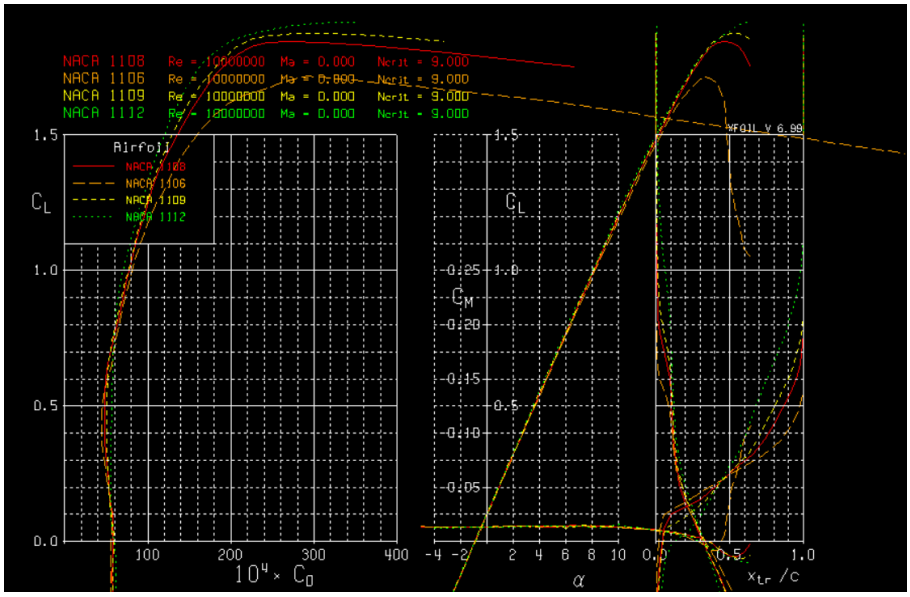


Fig. 12. Thickness variation for NACA 11xx

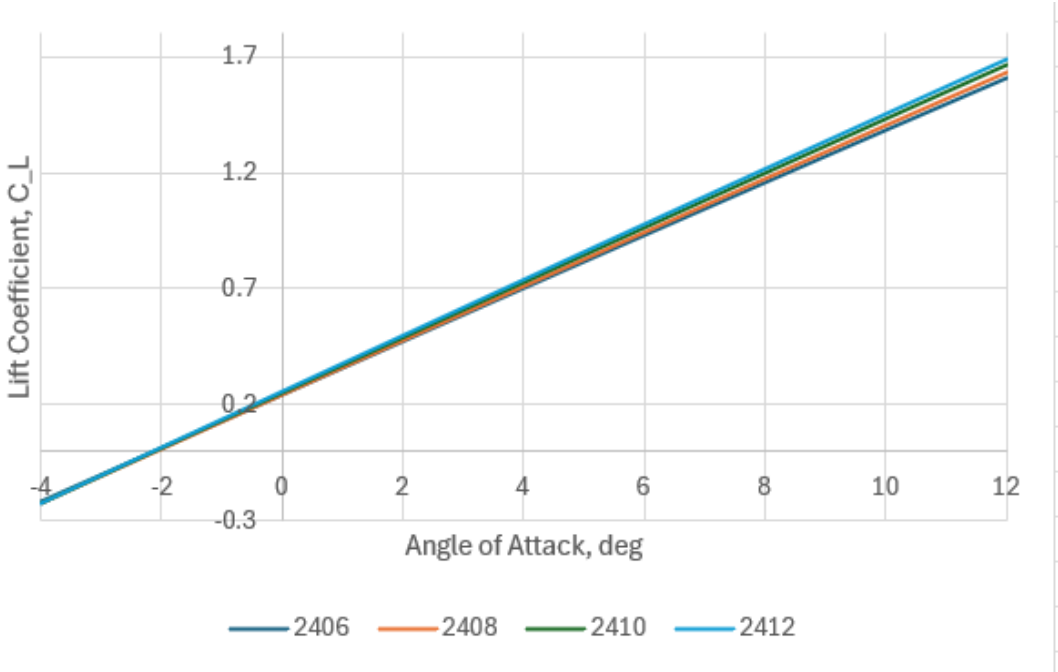


Fig. 13. Effect of Thickness Variation on the Inviscid Lift Curves of the NACA 24xx Airfoil Family

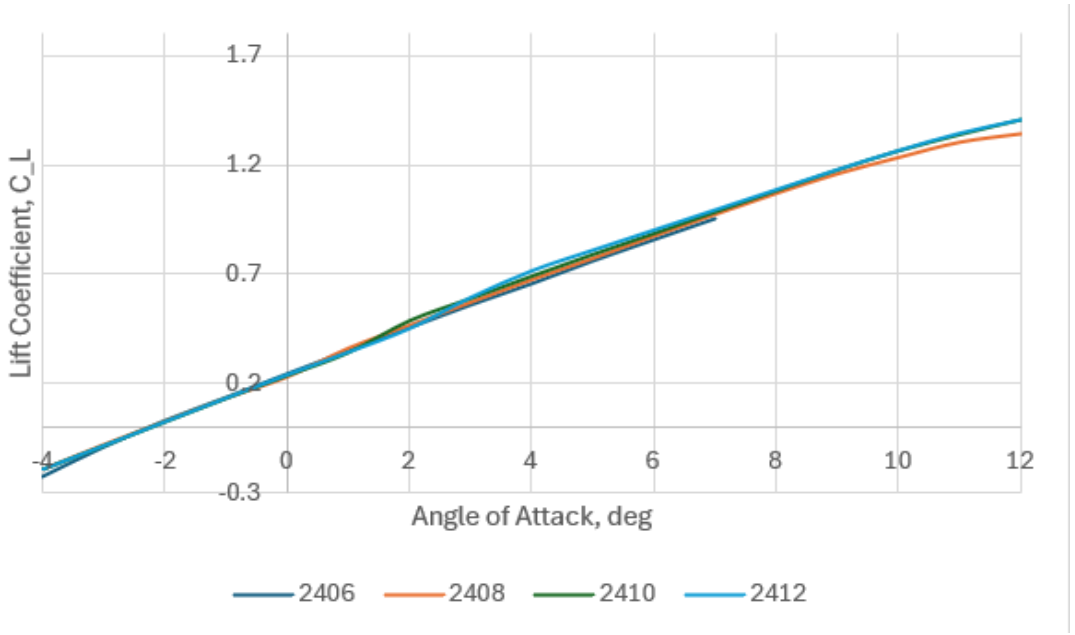


Fig. 14. Effect of Thickness Variation on the Viscous Lift Curves of the NACA 24xx Airfoil Family

G. Pressure-Coefficient Distributions

Figures 15-29 show the pressure coefficient (C_p) distributions as a function of chord position (x/c) for the airfoils using both viscous and inviscid solutions at various angles of attack. All the cases showed that the inviscid solutions at every angle of attack chosen displayed a slightly increased coefficient of pressure over the upper and lower side of the airfoil. At higher angles of attack, the viscous cases show a reduced suction peak and a more gradual pressure recovery, which can be seen most prominently along the upper surface of the airfoil. This trend occurs because the presence of viscosity introduces boundary layer effects, which reduces the local velocities near the surface and promotes earlier flow separation compared to the inviscid ideal case.

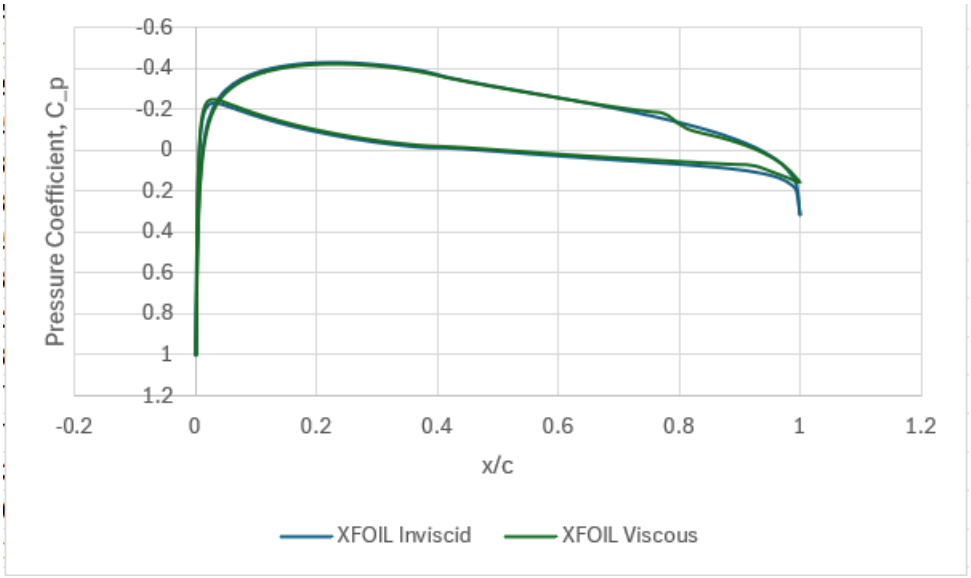


Fig. 15. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 2408 at α=0°

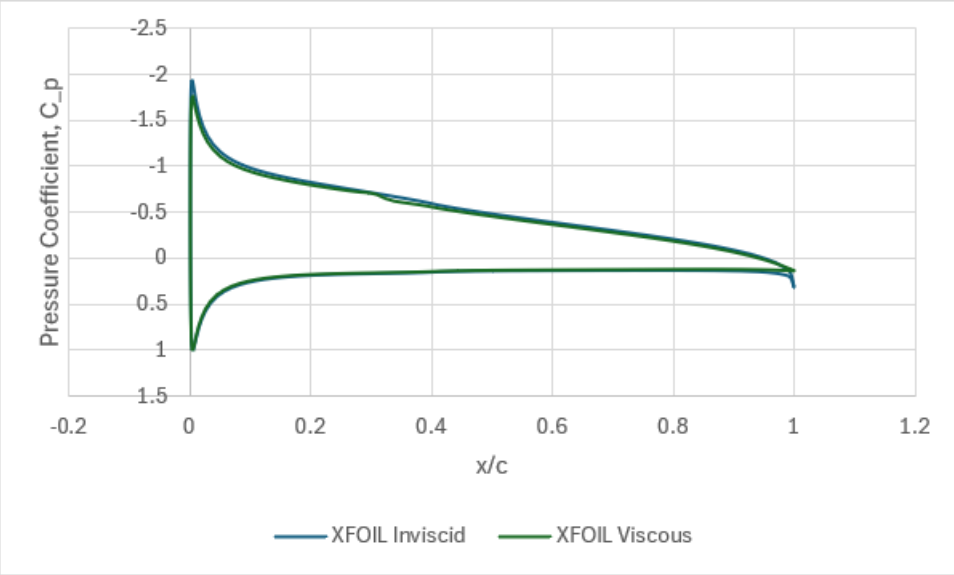


Fig. 16. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 2408 at α=4°

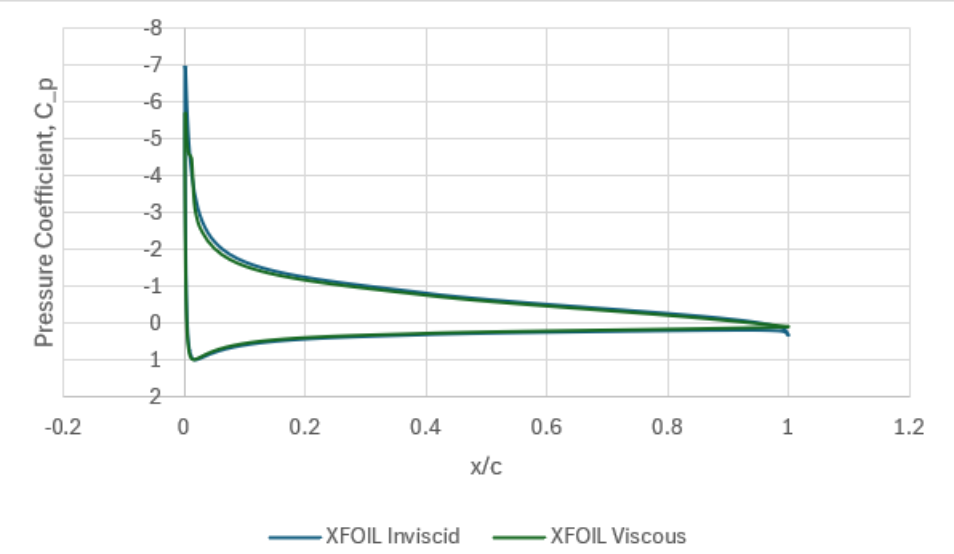


Fig. 17. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 2408 at α=8°

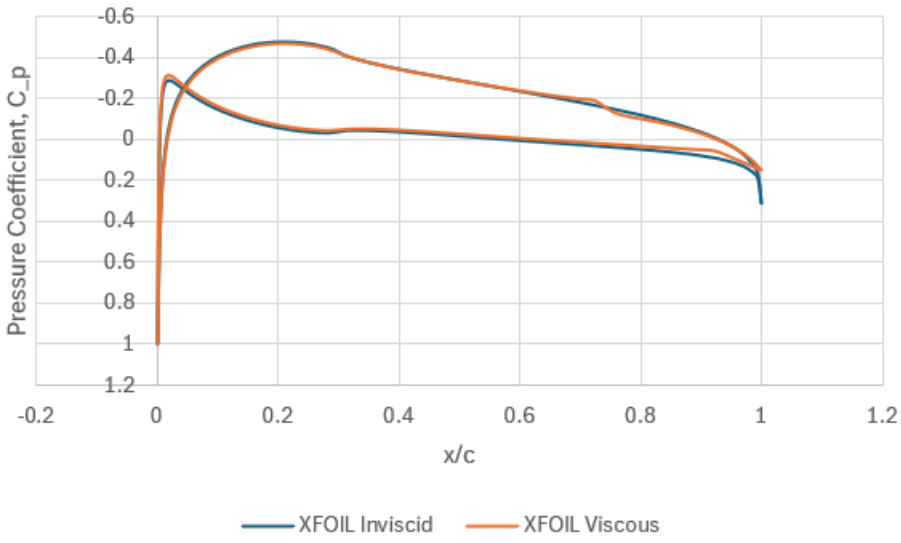


Fig. 18. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 2308 at α=0°

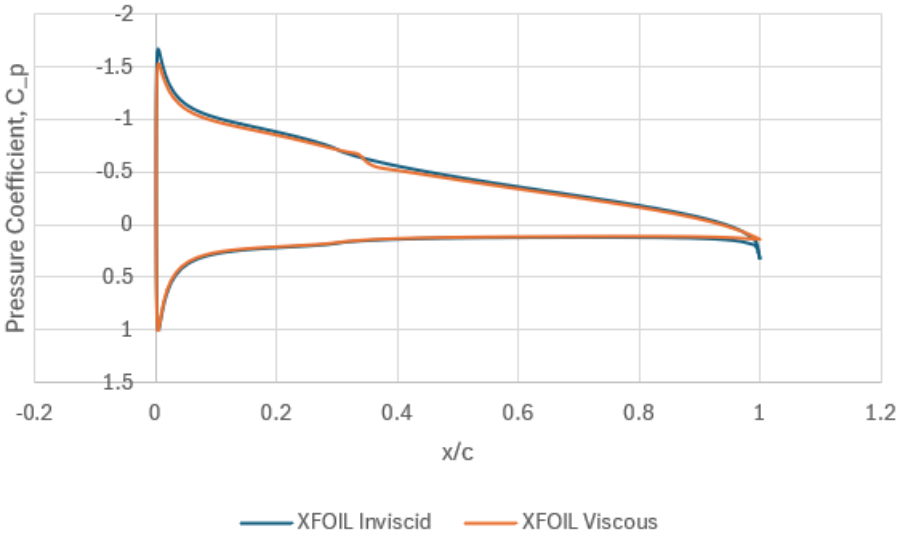


Fig. 19. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 2308 at α=4°

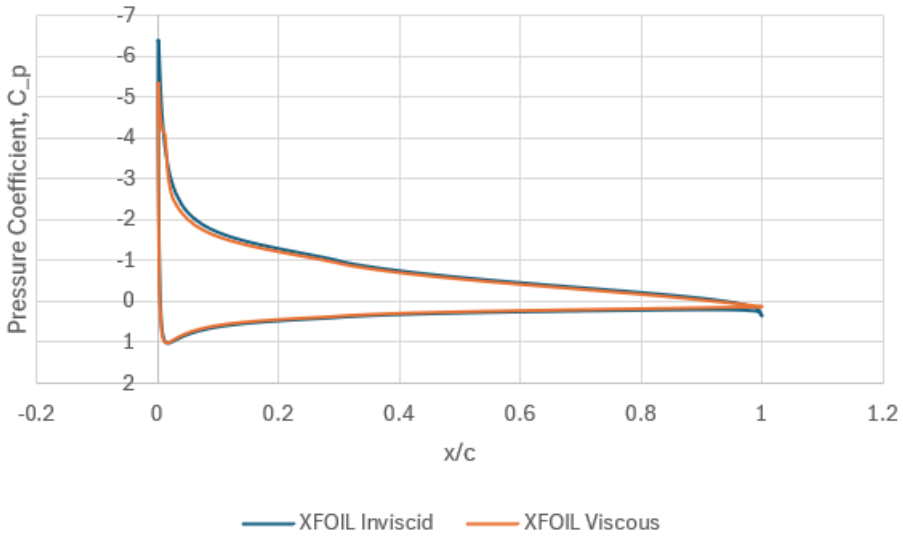


Fig. 20. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 2308 at α=8°

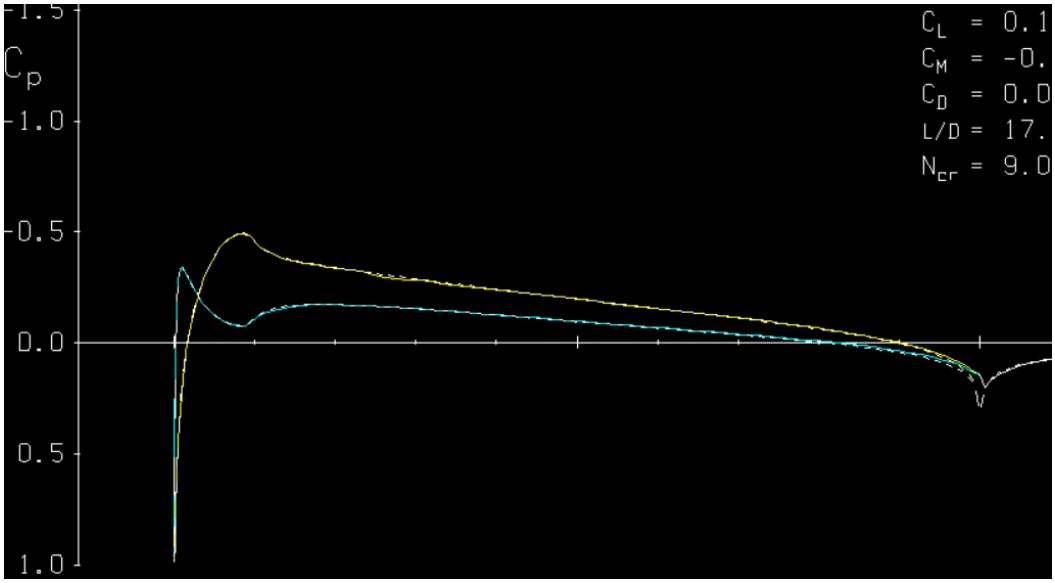


Fig. 21. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 1108 at α=0°

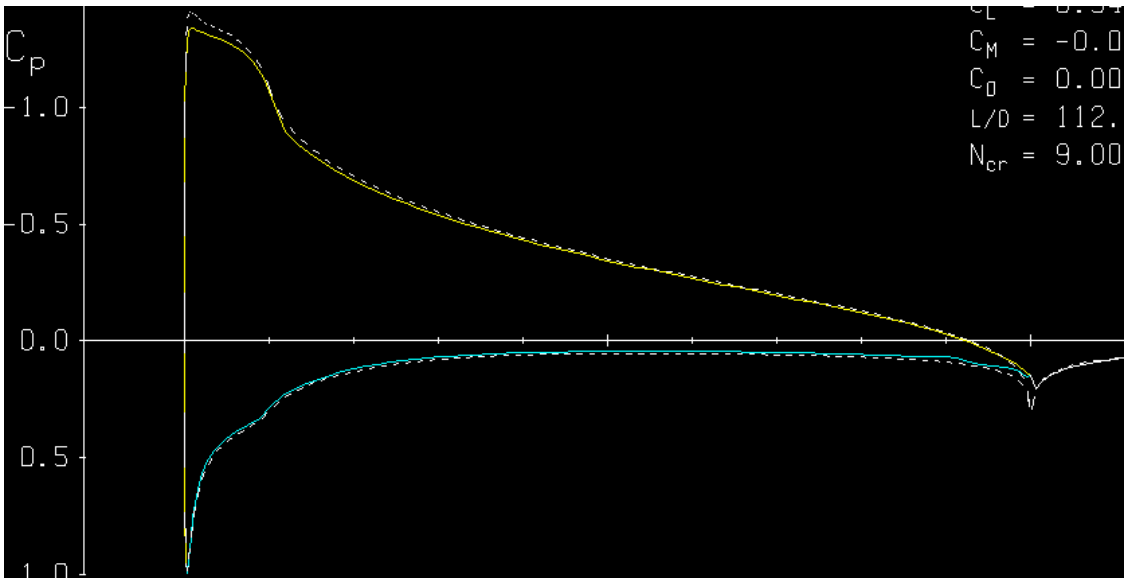


Fig. 22. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 1108 at α=4º

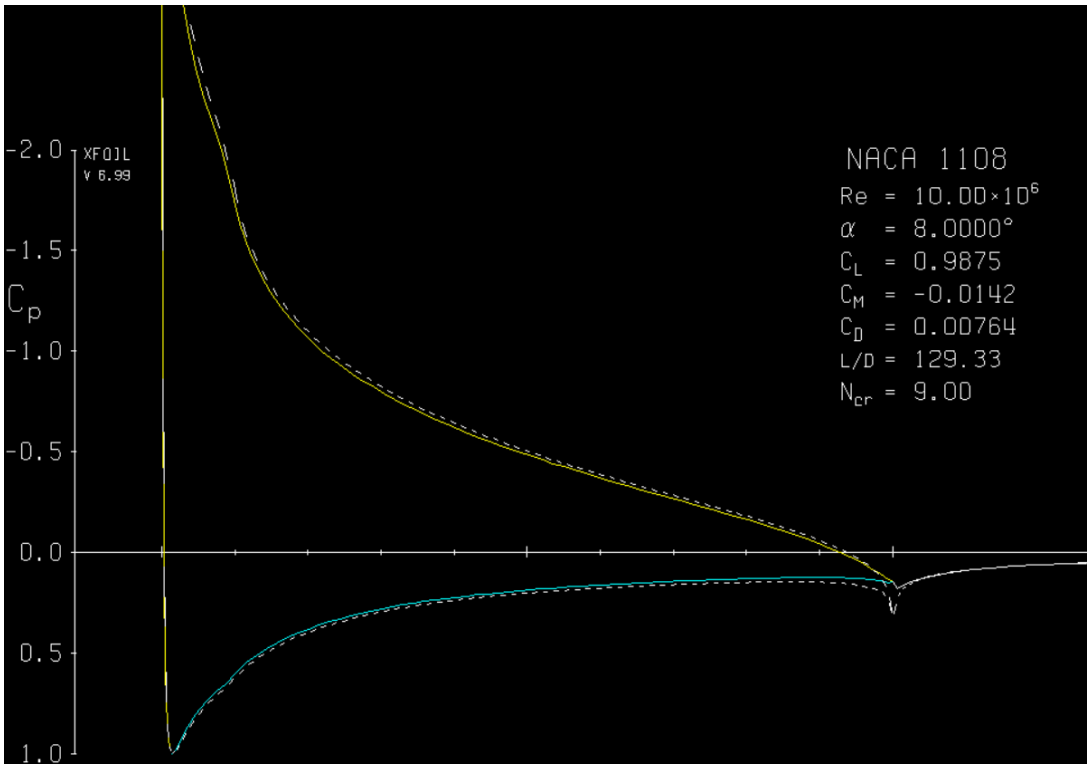


Fig. 23. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 1108 at α=8º

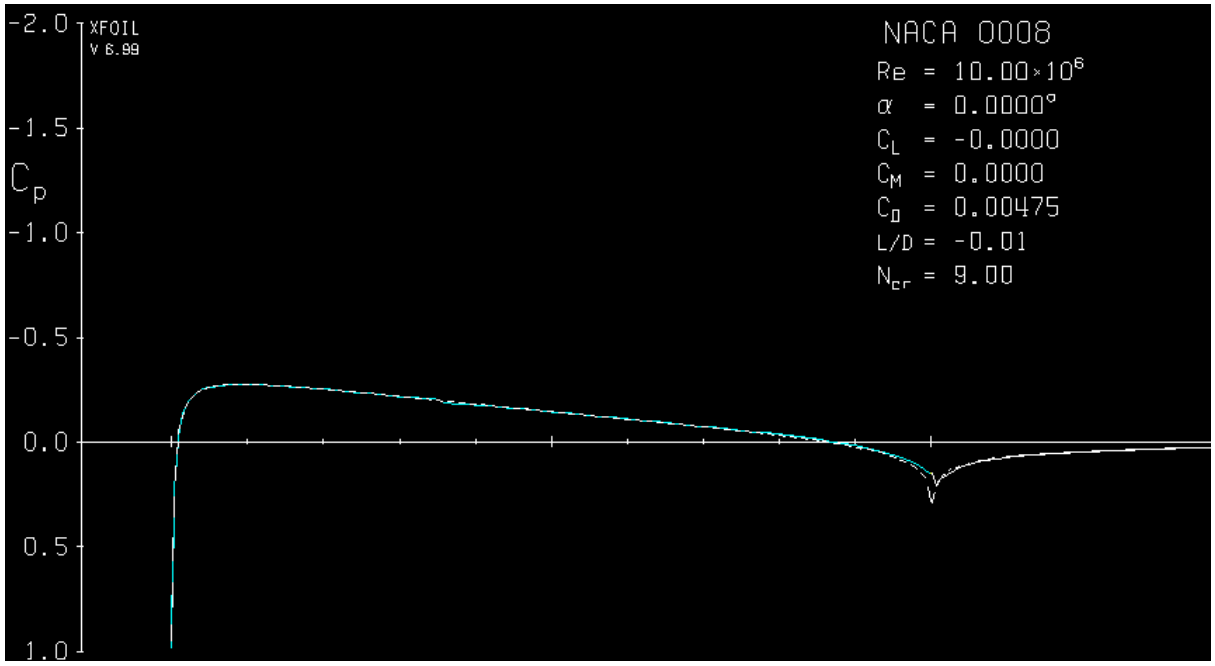


Fig. 24. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 0008 at α=0º

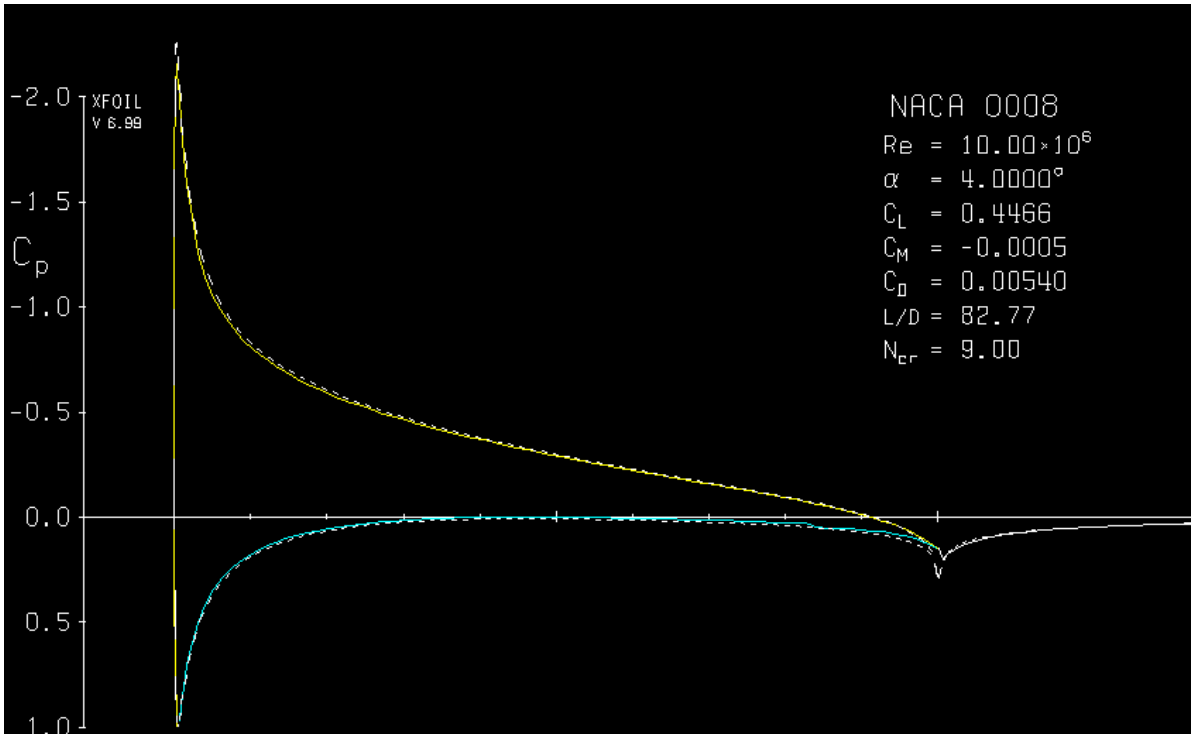


Fig. 25. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 0008 at α=4º

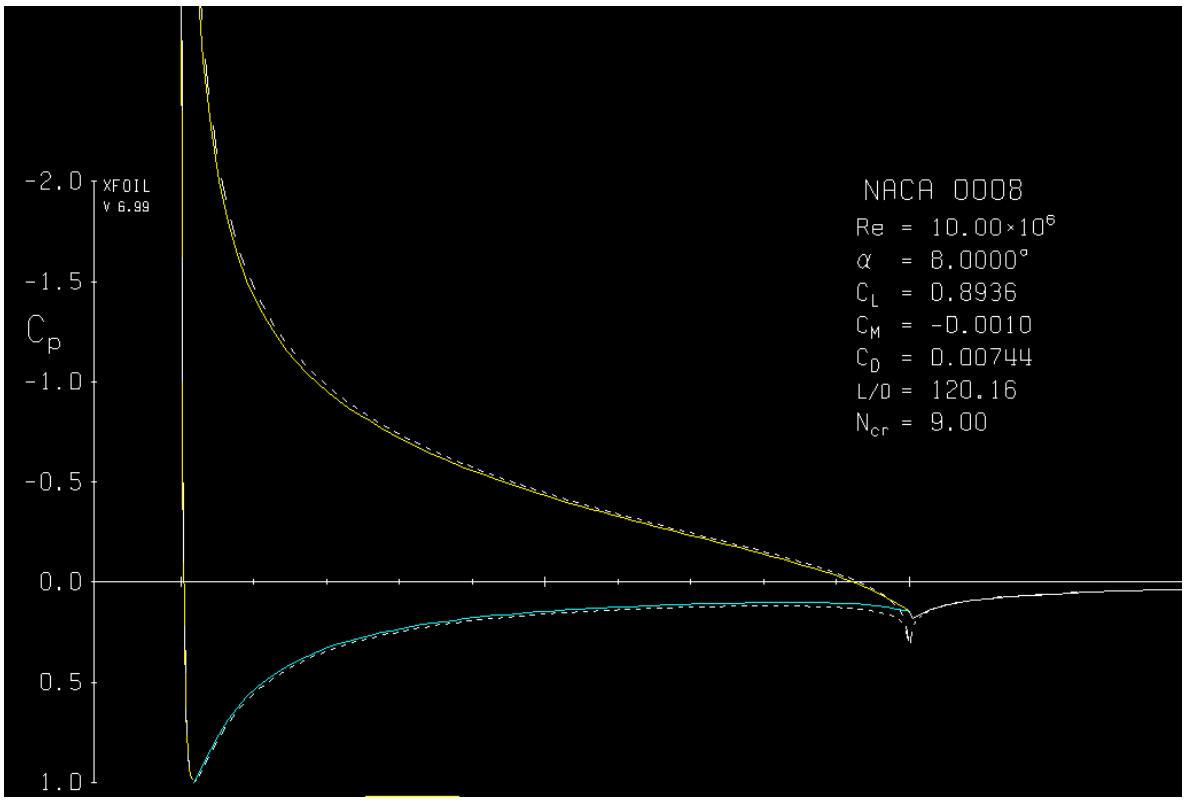


Fig. 26. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 0008 at α=8º

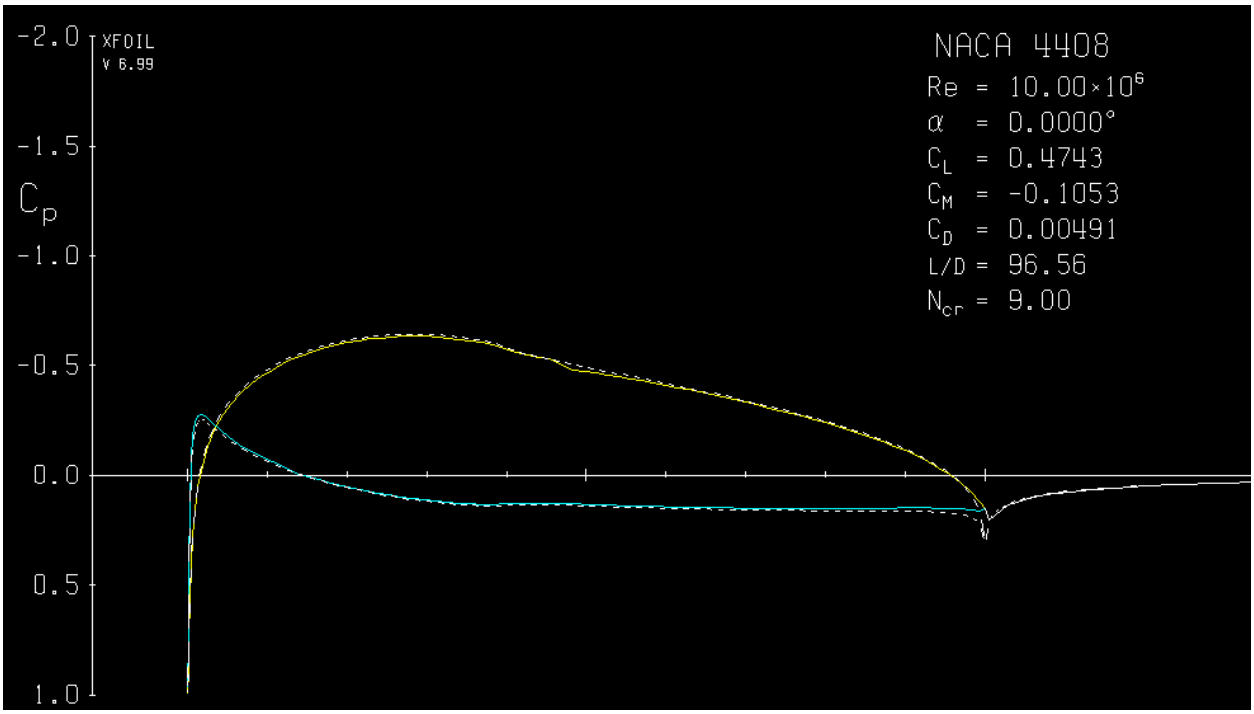


Fig. 27. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 4408 at α=0º

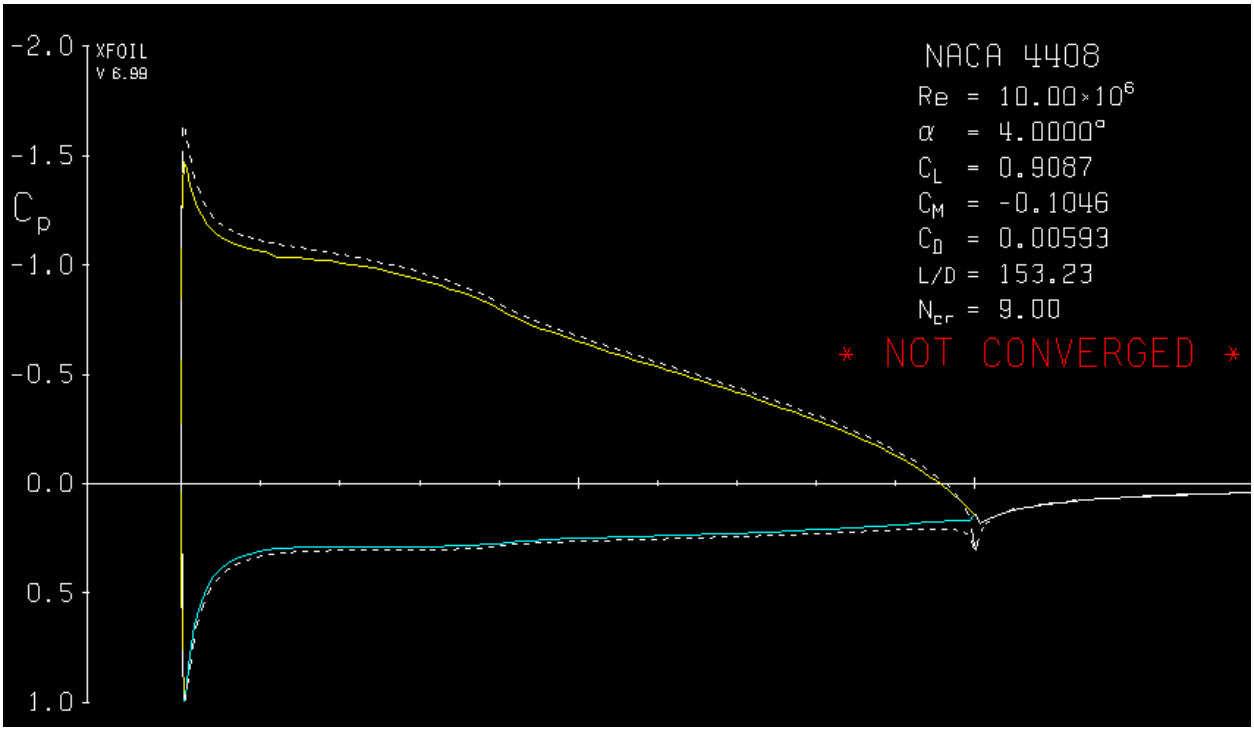


Fig. 28. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 4408 at α=4º

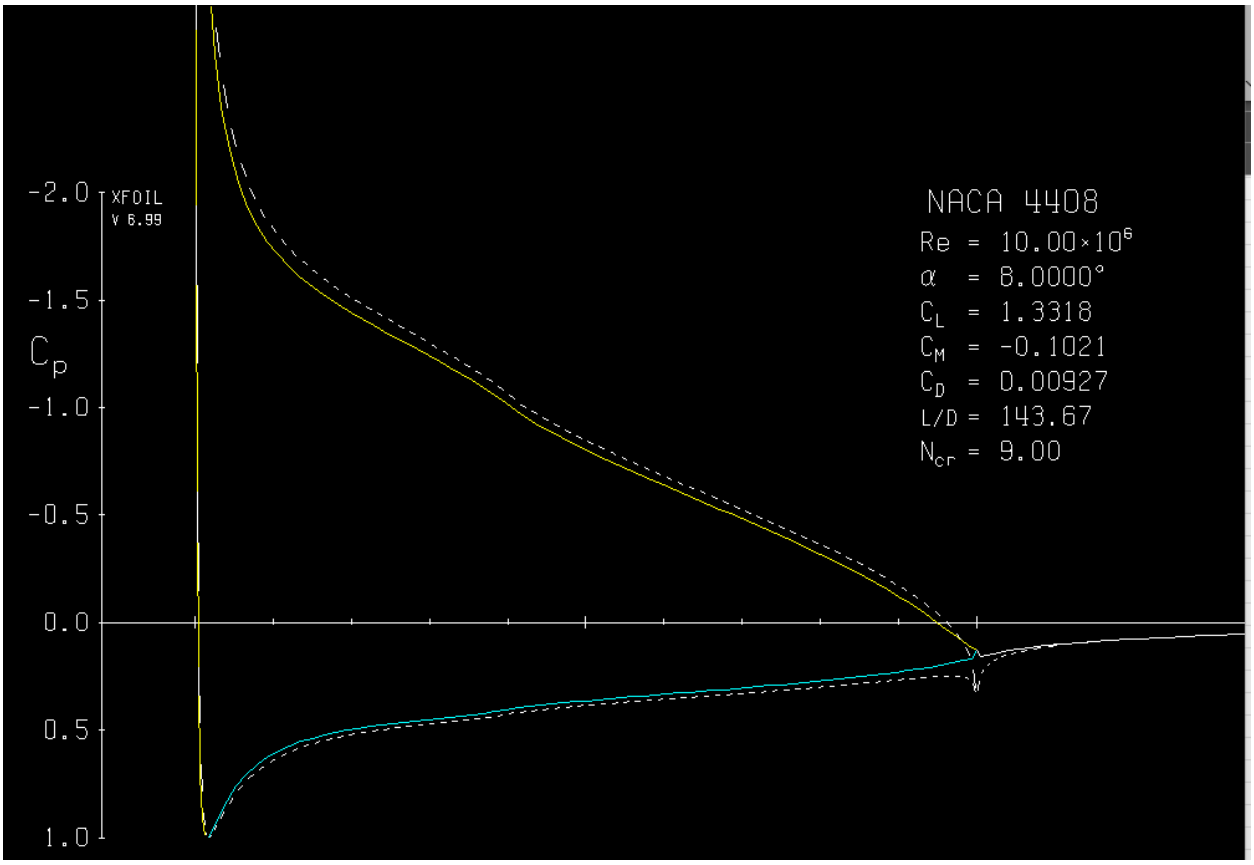


Fig. 29. Inviscid and Viscous Pressure-Coefficient Distributions for NACA 4408 at α=8º

IV. Discussion

The results of this study demonstrate clear trends among airfoil geometry, flow conditions, and lift behavior, even though there were some discrepancies between methods. When comparing this airfoil theory (TAT) to XFOIL, it could be seen that both methods agreed well in the lower angles of attack and started to deviate around the higher angles of attack. This is most likely since TAT assumes a simple model of the airfoil and doesn't account for thickness and the actual geometry of the airfoil, like XFOIL does using the panel method. XFOIL, using a panel method, gives a more detailed solution when analyzing an airfoil. This shows up even in the inviscid cases; there are slight differences because of how the airfoil is represented mathematically.

Looking at camber effects, it becomes obvious that increasing camber increases the lift and shifts the angle of zero lift further into the negative direction. It can also be seen that airfoils with more forward camber seem to produce lift more aggressively at lower angles of attack, which makes sense since the pressure distribution is being affected earlier along the chord length. The NACA 0008 case behaved exactly as a symmetric airfoil should, with zero lift at a zero angle of attack, which acted as a great kind of baseline to measure the accuracy of the data.

The inviscid versus viscous comparisons showed more differences, obviously, specifically around higher angles of attack, where the viscous cases started to round off and peak. The main reason for this is because of boundary layer effects and flow separation. These effects cause a reduction in lift in real flow conditions, which is why the inviscid case just keeps going like nothing is affecting it, because there isn't any.

The Reynolds number variation study also showed a consistent trend where increasing the Reynolds number improved the lift performance and delayed flow separation. At the lower Reynolds numbers tested, the airfoil struggled to maintain lift at higher angles of attack, which is depicted in the graph when the curves start to bend early, while higher Reynolds numbers kept a linear increase for longer. This is mostly because at higher Reynolds numbers, there is a reduction in the relative thickness of the boundary layer, which keeps the flow attached for longer. There was also an increase in the maximum lift for the airfoil with the increase in the Reynolds number.

When analyzing the effects of thickness on an airfoil, it was observed that an increase in thickness causes a slight increase in the lift as well. From the data, it can be seen that the most prominent effect of a variation of thickness is the steepness of the stall point slope. The thicker the airfoil is, the more gradual the lift slope comes to its stall point. The highest thickness tested (12%) had the highest coefficient of lift (Cl) and the least steep curve into the point of stall once the airfoil deviated from its linear lift increase.

V. Conclusion

In conclusion, this study looked at and analyzed the aerodynamic behavior of several NACA 4-digit airfoils using both thin airfoil theory (TAT) and XFOIL, which operates using panel methods. The results showed that TAT works well for predicting lift in the linear region, especially around lower angles of attack. However, once viscosity is introduced, making the conditions more realistic and higher angles of attack, the differences between thin airfoil theory (TAT) and XFOIL become more apparent.

Overall, the results showed that both airfoil geometry and real flow conditions both have major roles in determining the aerodynamic performance of an airfoil, and even small changes can have noticeable effects on lift behavior. While the general trends followed what was expected from theory, the XFOIL analysis helped show where those assumptions start to break down. This analysis made it clear that using multiple methods together can give an overall better understanding than just relying on one, since each approach captures different aspects of flow, even if they aren't completely similar.

VI. Appendix

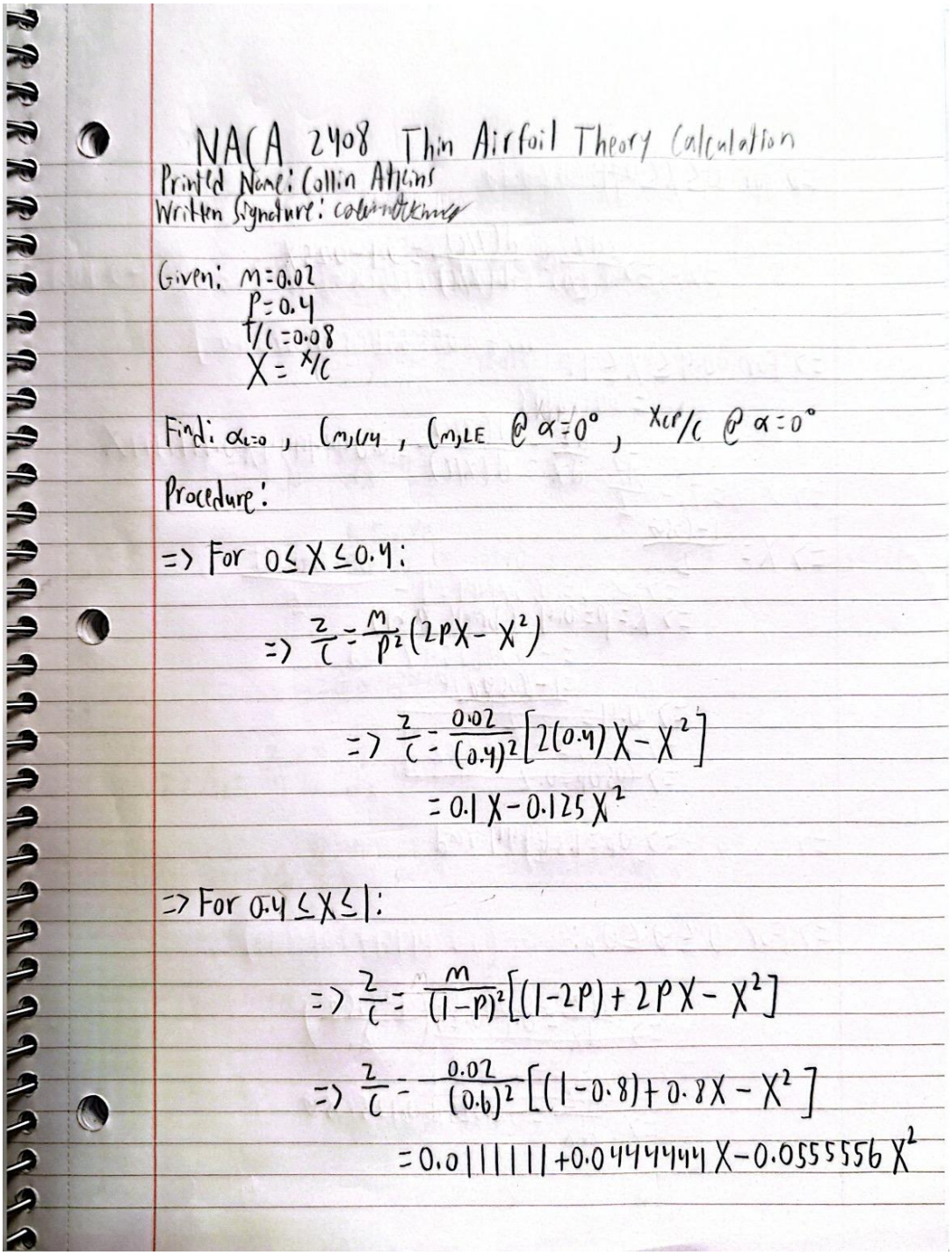


Fig. A1. Handwritten Thin Airfoil Theory Calculation for NACA 2408, Page 1

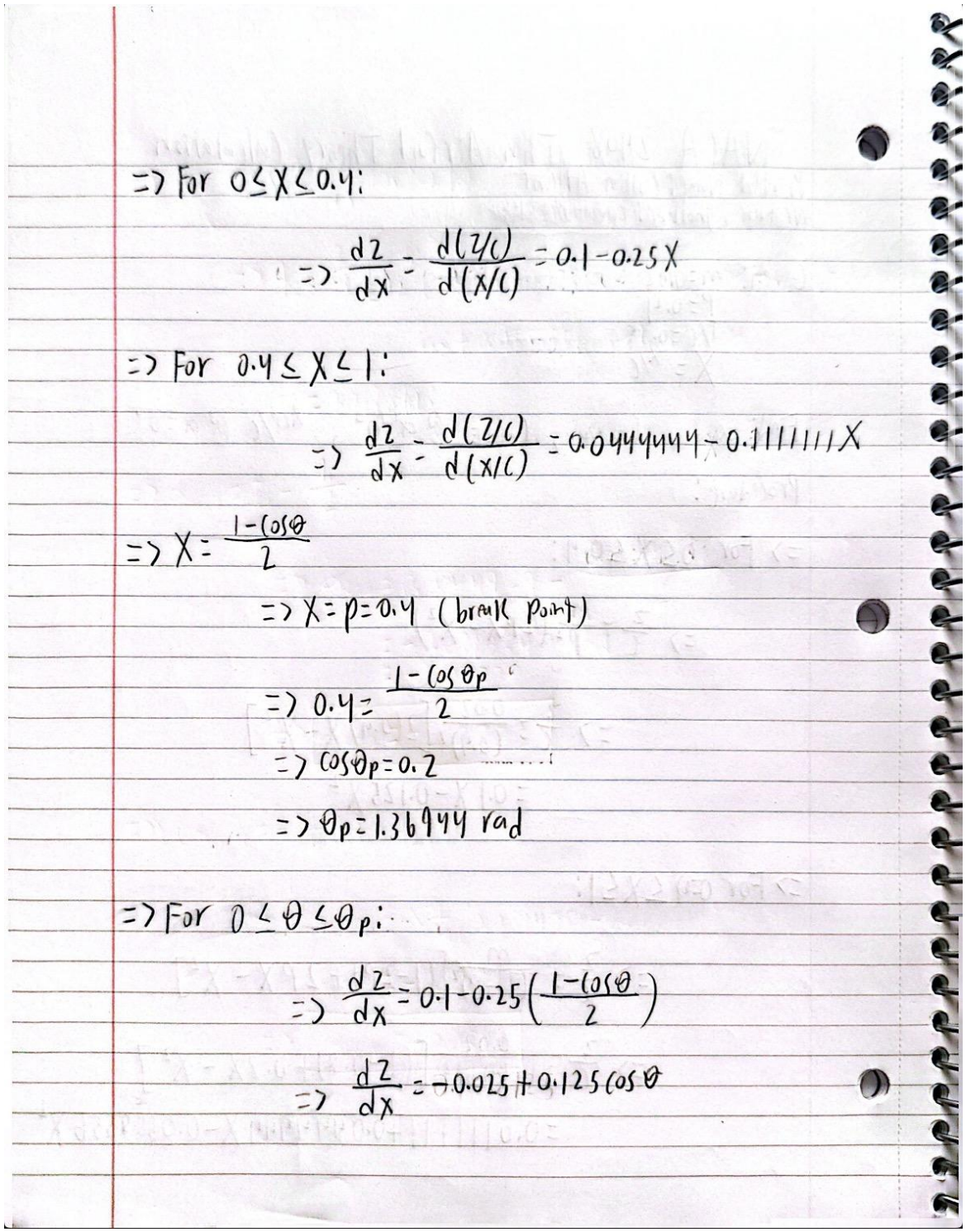


Fig. A2. Handwritten Thin Airfoil Theory Calculation for NACA 2408, Page 2

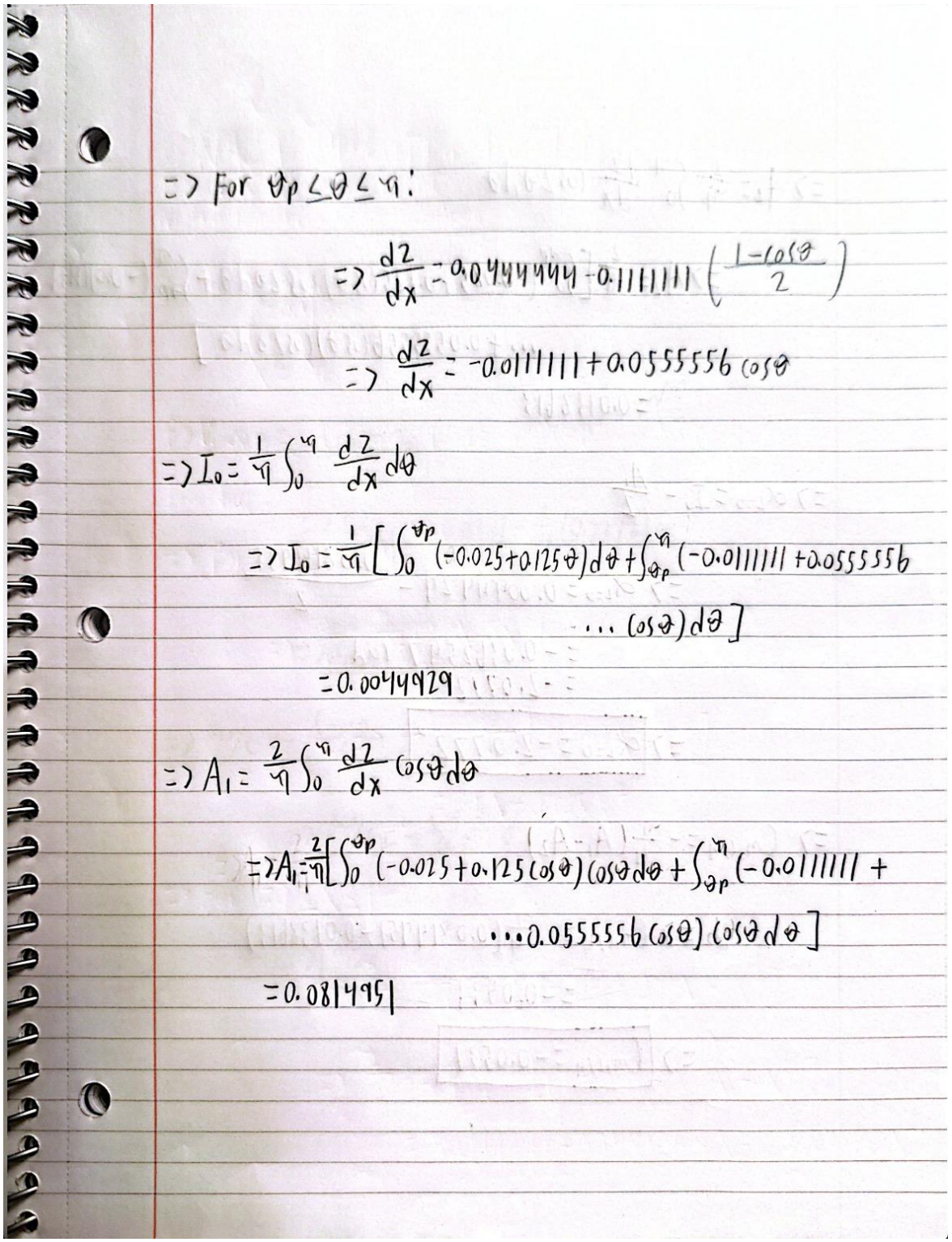


Fig. A3. Handwritten Thin Airfoil Theory Calculation for NACA 2408, Page 3

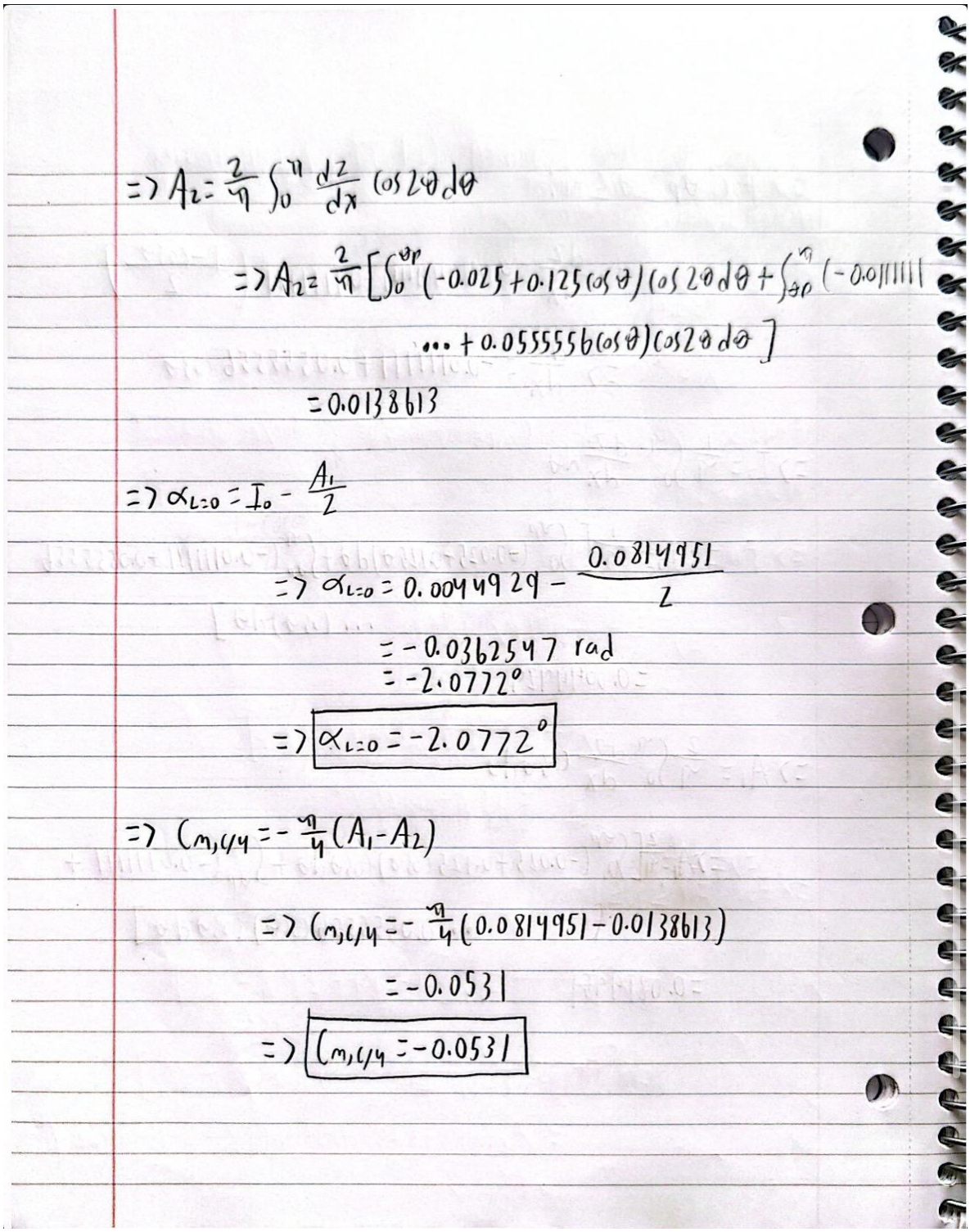


Fig. A4. Handwritten Thin Airfoil Theory Calculation for NACA 2408, Page 4

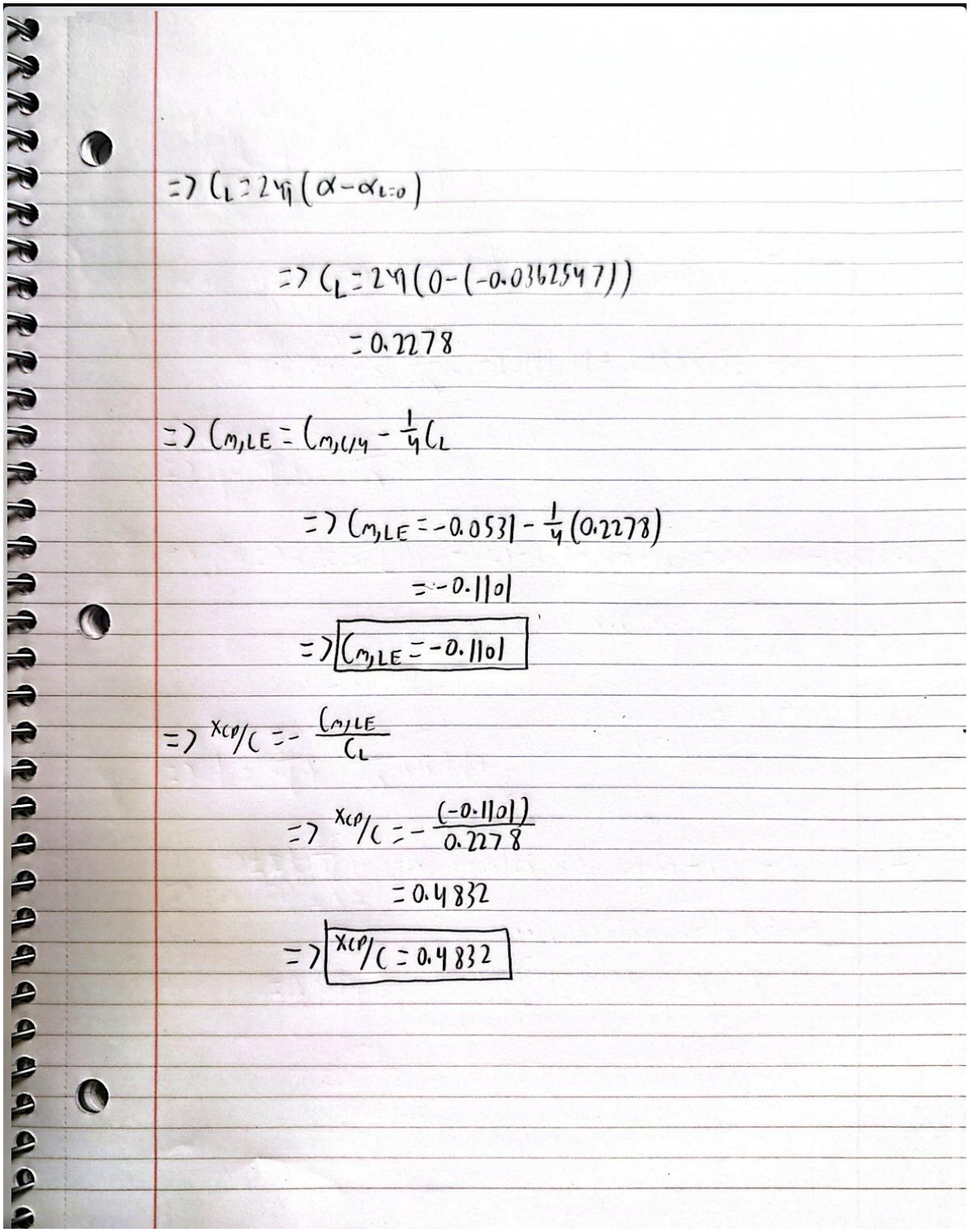


Fig. A5. Handwritten Thin Airfoil Theory Calculation for NACA 2408, Page 5

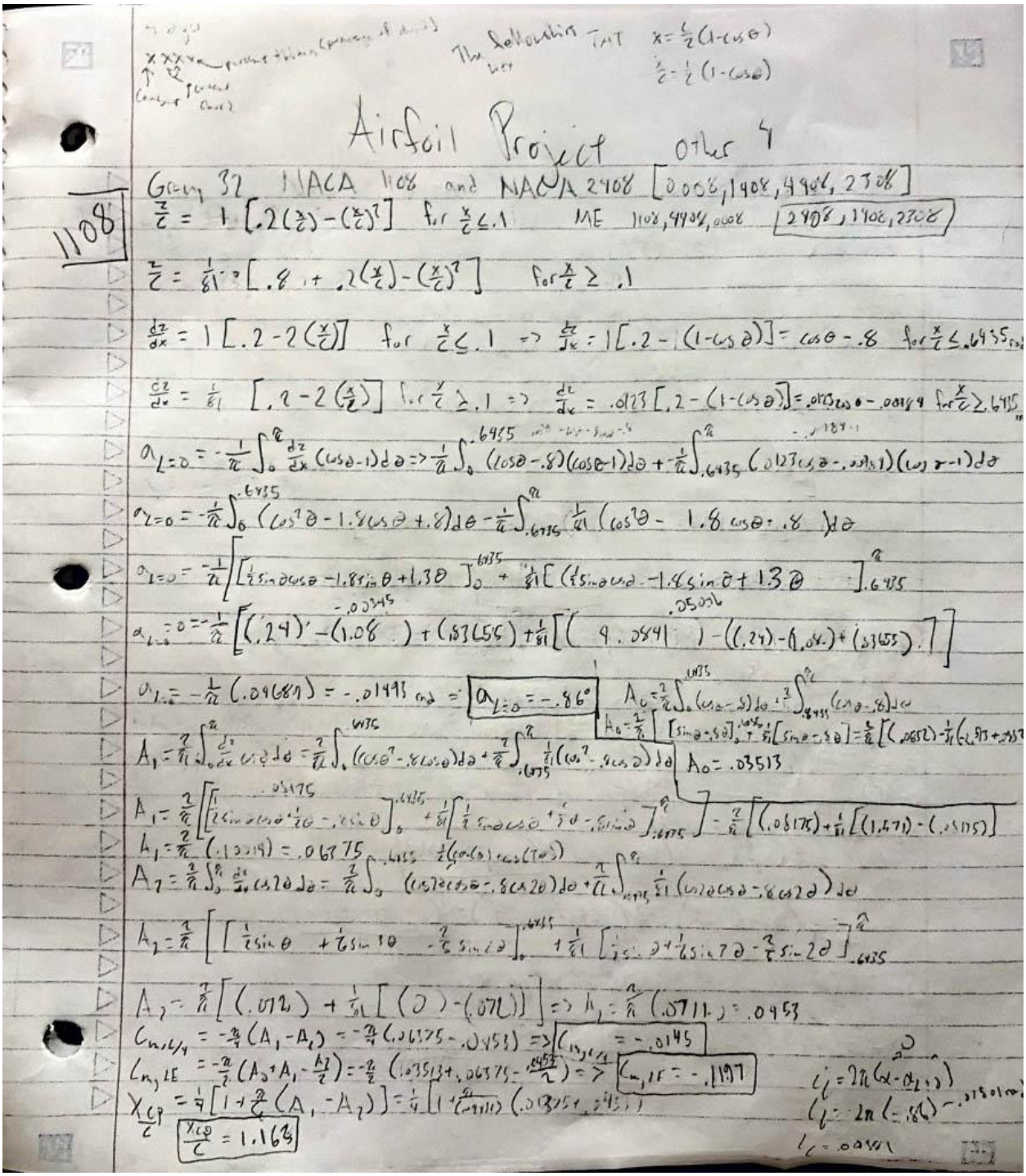


Fig. A6. Handwritten Thin Airfoil Theory Calculation for NACA 1108, Page 1

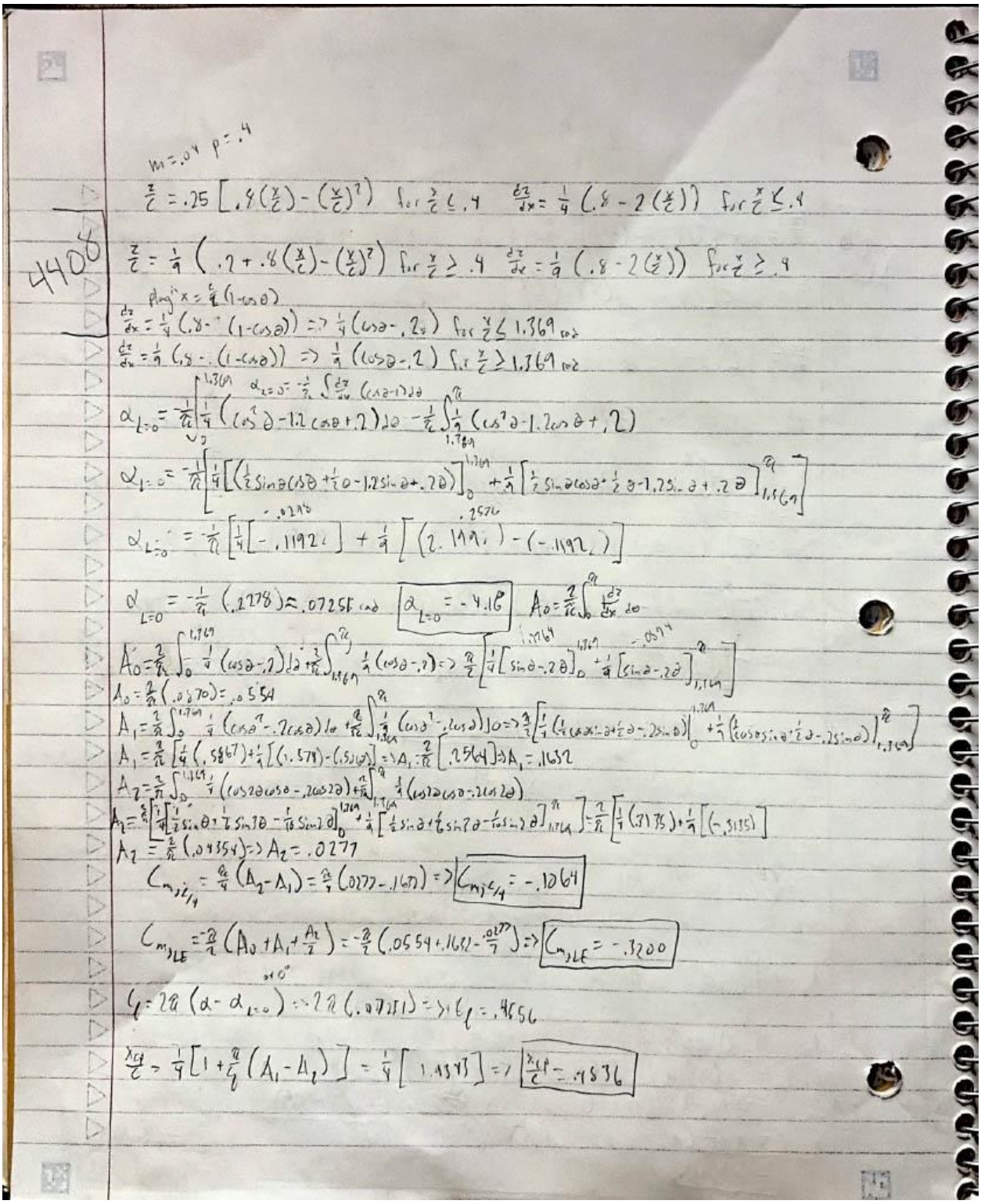


Fig. A7. Handwritten Thin Airfoil Theory Calculation for NACA 4408, Page 1

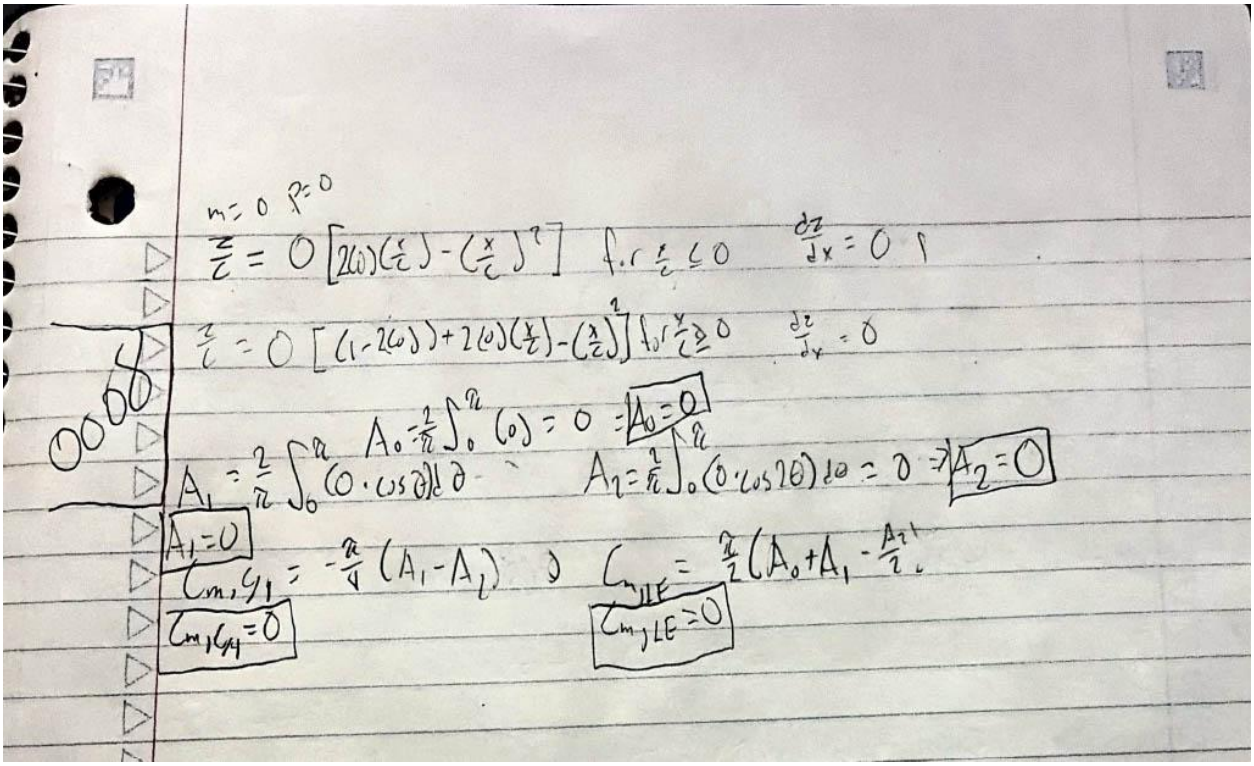


Fig. A8. Handwritten Thin Airfoil Theory Calculation for NACA 0008, Page 1

	A	B	C	D	E	F	G
1	x/c	2408	1408	2308	1108	4408	8
2	0	0	0	0	0	0	0
3	0.05	0.004688	0.002344	0.006111	0.0075	0.009375	0
4	0.1	0.00875	0.004375	0.011111	0.01	0.0175	0
5	0.15	0.012188	0.006094	0.015	0.009969	0.024375	0
6	0.2	0.015	0.0075	0.017778	0.009877	0.03	0
7	0.25	0.017188	0.008594	0.019444	0.009722	0.034375	0
8	0.3	0.01875	0.009375	0.02	0.009506	0.0375	0
9	0.35	0.019688	0.009844	0.019898	0.009228	0.039375	0
10	0.4	0.02	0.01	0.019592	0.008889	0.04	0
11	0.45	0.019861	0.009931	0.019082	0.008488	0.039722	0
12	0.5	0.019444	0.009722	0.018367	0.008025	0.038889	0
13	0.55	0.01875	0.009375	0.017449	0.0075	0.0375	0
14	0.6	0.017778	0.008889	0.016327	0.006914	0.035556	0
15	0.65	0.016528	0.008264	0.015	0.006265	0.033056	0
16	0.7	0.015	0.0075	0.013469	0.005556	0.03	0
17	0.75	0.013194	0.006597	0.011735	0.004784	0.026389	0
18	0.8	0.011111	0.005556	0.009796	0.003951	0.022222	0
19	0.85	0.00875	0.004375	0.007653	0.003056	0.0175	0
20	0.9	0.006111	0.003056	0.005306	0.002099	0.012222	0
21	0.95	0.003194	0.001597	0.002755	0.00108	0.006389	0
22	1	-1E-07	-4E-08	-1.4E-17	-5.2E-18	-4.2E-17	0

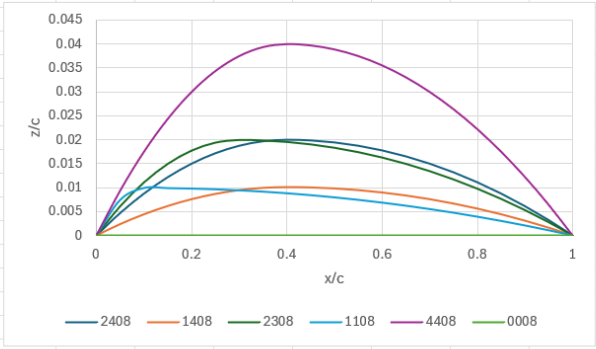


Fig. A9. EXCEL Mean Camber Lines

	A	B	C	D
1	alpha (deg)	TAT CL	XFOIL inviscid CL	XFOIL viscous CL
2	-4	-0.21086	-0.2217	-0.1943
3	-3	-0.1012	-0.1045	-0.0808
4	-2	0.008466	0.0126	0.0263
5	-1	0.118128	0.1298	0.1325
6	0	0.22779	0.2469	0.2294
7	1	0.337453	0.3639	0.3605
8	2	0.447115	0.4808	0.4654
9	3	0.556777	0.5976	0.5709
10	4	0.66644	0.7142	0.6747
11	5	0.776102	0.8306	0.772
12	6	0.885764	0.9467	0.8743
13	7	0.995426	1.0625	0.9732
14	8	1.105089	1.178	1.0695
15	9	1.214751	1.2932	1.1591
16	10	1.324413	1.408	1.2337
17	11	1.434075	1.5223	1.3036
18	12	1.543738	1.6362	1.342
19				

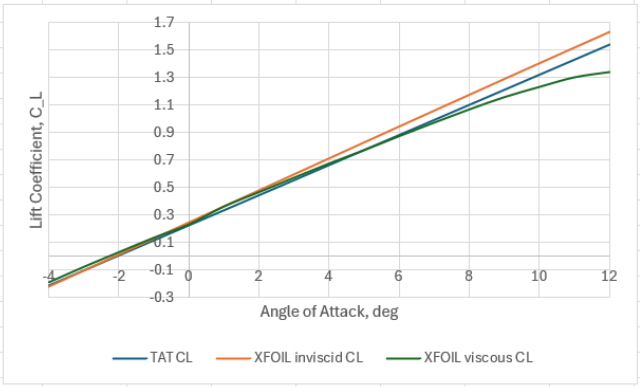


Fig. A10. EXCEL NACA 2408 Lift-Curve Comparison

	A	B	C	D
1	alpha deg	TAT CL	XFOIL inviscid CL	XFOIL viscous CL
2	-4	-0.32475	-0.3448	-0.3419
3	-3	-0.21509	-0.2278	-0.2056
4	-2	-0.10543	-0.1107	-0.0874
5	-1	0.004233	0.0064	0.0156
6	0	0.113895	0.1235	0.1143
7	1	0.223558	0.2405	0.2335
8	2	0.33322	0.3575	0.3639
9	3	0.442882	0.4744	0.4649
10	4	0.552544	0.5911	0.5629
11	5	0.662207	0.7077	0.6642
12	6	0.771869	0.824	0.7676
13	7	0.881531	0.9401	0.8668
14	8	0.991193	1.0559	0.9671
15	9	1.100856	1.1714	1.0557
16	10	1.210518	1.2865	1.1412
17	11	1.32018	1.4012	1.1882
18	12	1.429842	1.5156	1.2269
19				
20				

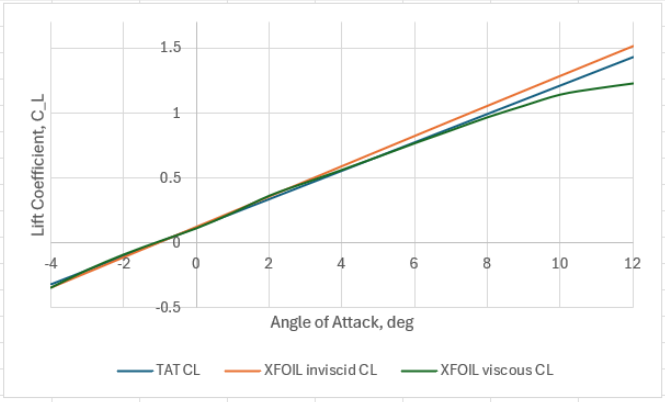


Fig. A11. EXCEL NACA 1408 Lift-Curve Comparison

	A	B	C	D
1	alpha (deg)	TAT CL	XFOIL invicid CL	XFOIL viscous CL
2	-4	-0.22832	-0.2399	-0.2107
3	-3	-0.11865	-0.1228	-0.0969
4	-2	-0.00899	-0.0057	0.0098
5	-1	0.10067	0.1115	0.1149
6	0	0.210332	0.2286	0.21
7	1	0.319995	0.3457	0.347
8	2	0.429657	0.4626	0.4505
9	3	0.539319	0.5794	0.556
10	4	0.648981	0.696	0.6628
11	5	0.758644	0.8124	0.7663
12	6	0.868306	0.9286	0.8639
13	7	0.977968	1.0445	0.9655
14	8	1.08763	1.16	1.0638
15	9	1.197293	1.2753	1.1534
16	10	1.306955	1.3901	1.2296
17	11	1.416617	1.5045	1.3031
18	12	1.526279	1.6184	1.3514
19				

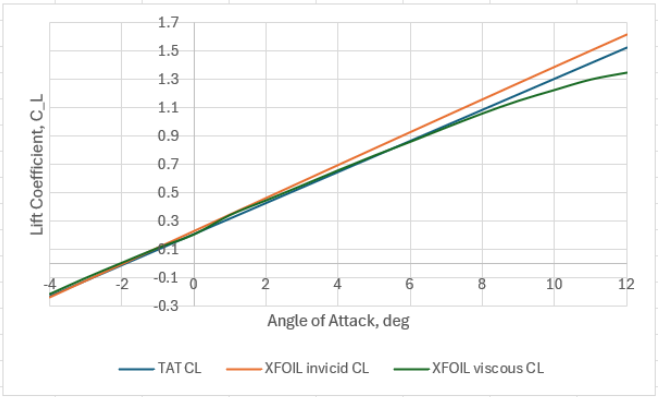


Fig. A12. EXCEL NACA 2308 Lift-Curve Comparison

	A	B	C	D	E
1	alpha	2406	2408	2410	2412
2	-4	-0.2278	-0.1943	-0.1931	-0.197
3	-3	-0.0921	-0.0808	-0.0851	-0.0877
4	-2	0.0289	0.0263	0.0236	0.0219
5	-1	0.1353	0.1325	0.1313	0.1304
6	0	0.2437	0.2294	0.2359	0.2372
7	1	0.3501	0.3605	0.3405	0.3416
8	2	0.4555	0.4654	0.4861	0.45
9	3	0.5601	0.5709	0.5874	0.5921
10	4	0.6578	0.6747	0.6895	0.7148
11	5	0.7623	0.772	0.7888	0.8095
12	6	0.8599	0.8743	0.8848	0.902
13	7	0.9565	0.9732	0.9829	0.9946
14	8		1.0695	1.0804	1.0874
15	9		1.1591	1.1749	1.1796
16	10		1.2337	1.2624	1.2672
17	11		1.3036	1.3366	1.3464
18	12		1.342	1.4044	1.4085
19					
20					

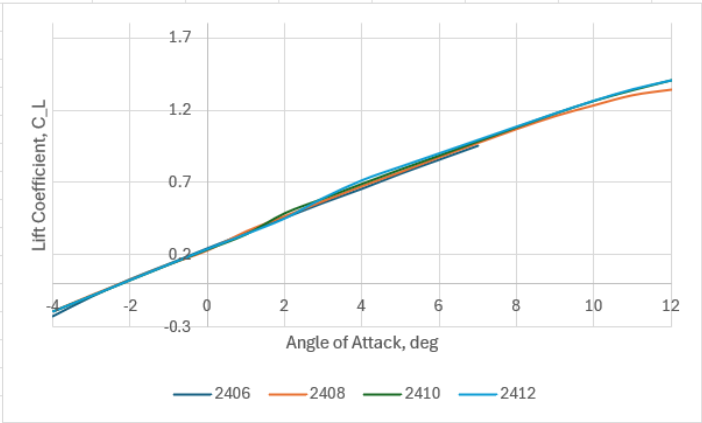


Fig. A13. EXCEL NACA 2408 Viscous Thickness Variation

G	H	I	J
x/c inv	Cp inv	x/c visc	Cp visc
1	0.32029	1	0.13315
0.99431	0.16693	0.99431	0.11737
0.98332	0.10326	0.98332	0.08722
0.9702	0.05412	0.9702	0.05521
0.9562	0.01697	0.9562	0.02507
0.94195	-0.01336	0.94195	-0.00228
0.92763	-0.04099	0.92763	-0.02759
0.91328	-0.06392	0.91328	-0.05026
0.89893	-0.08548	0.89893	-0.07142
0.88457	-0.10566	0.88457	-0.09128
0.8702	-0.12401	0.8702	-0.10978
0.85583	-0.14199	0.85583	-0.1275
0.84144	-0.15813	0.84144	-0.144
0.82706	-0.17415	0.82706	-0.16001
0.81266	-0.18943	0.81266	-0.1754
0.79827	-0.2047	0.79827	-0.19046
0.78386	-0.21882	0.78386	-0.20477
0.76946	-0.23277	0.76946	-0.21878
0.75505	-0.24699	0.75505	-0.23274
0.74063	-0.25982	0.74063	-0.24589
0.72621	-0.27381	0.72621	-0.25943
0.71179	-0.28622	0.71179	-0.27214
0.69737	-0.29955	0.69737	-0.28518
0.68294	-0.3125	0.68294	-0.29798
0.66851	-0.32517	0.66851	-0.31063
0.65408	-0.33843	0.65408	-0.32345

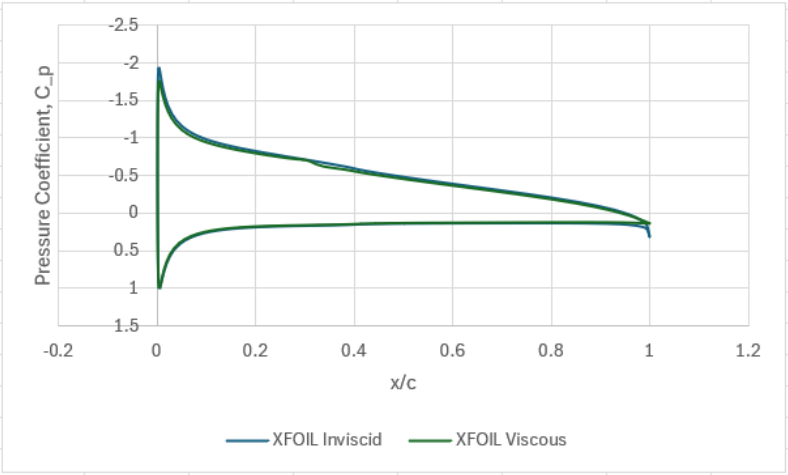


Fig. A14. EXCEL Pressure-Coefficient Distributions for NACA 2408 at α=4°

W	X	Y	Z	AA	AB	AC	AD	AE	AF
x/c inv	Cp inv	x/c visc	Cp visc						
1	0.3332	1	0.10183						
0.99431	0.17568	0.99431	0.09305						
0.98332	0.10553	0.98332	0.0753						
0.9702	0.05134	0.9702	0.05373						
0.9562	0.00867	0.9562	0.03055						
0.94195	-0.02613	0.94195	0.00707						
0.92763	-0.0589	0.92763	-0.01661						
0.91328	-0.08564	0.91328	-0.03965						
0.89893	-0.11169	0.89893	-0.06239						
0.88457	-0.13636	0.88457	-0.08475						
0.8702	-0.15832	0.8							
0.85583	-0.18124	0.85							
0.84144	-0.20105	0.84							
0.82706	-0.22125	0.82							
0.81266	-0.24071	0.81							
0.79827	-0.26034	0.79							
0.78386	-0.27867	0.78							
0.76946	-0.29662	0.76							
0.75505	-0.31531	0.75							
0.74063	-0.33225	0.74							
0.72621	-0.35093	0.72							
0.71179	-0.36762	0.71							
0.69737	-0.3855	0.69							
0.68294	-0.40329	0.68							
0.66851	-0.42037	0.66							
0.65408	-0.43909	0.65408	-0.39007						

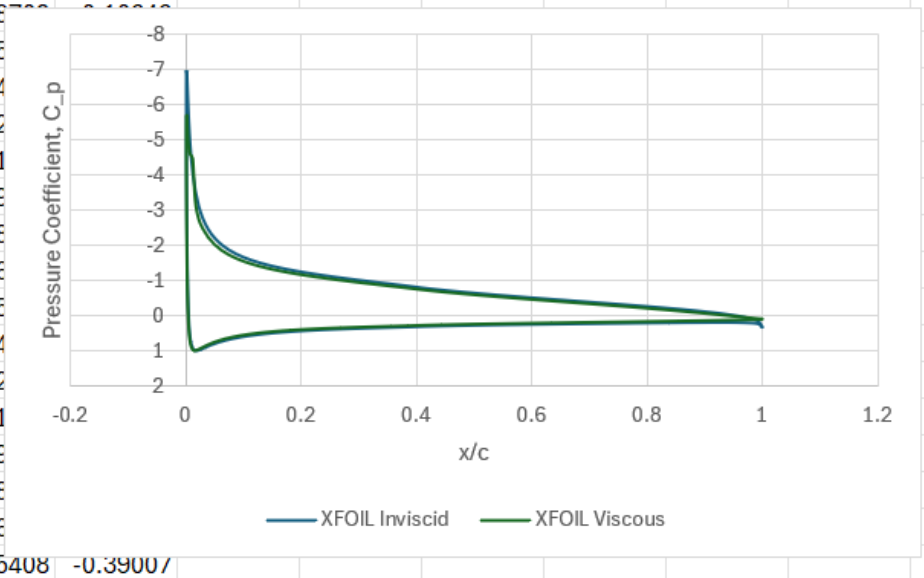


Fig. A15. EXCEL Pressure-Coefficient Distributions for NACA 2408 at α=8°

x/c	XFOIL Invis	XFOIL Viscous
1	0.31444	0.15186
0.99431	0.16862	0.13404
0.9833	0.11419	0.10046
0.97014	0.07338	0.0698
0.9561	0.04333	0.04385
0.94181	0.01909	0.02197
0.92746	-0.00181	0.00288
0.91308	-0.0195	-0.01359
0.89869	-0.03579	-0.0285
0.8843	-0.04988	-0.04152
0.8699	-0.06332	-0.0536
0.85549	-0.07644	-0.06498
0.84108	-0.08728	-0.07443
0.82667	-0.09908	-0.08393
0.81225	-0.10875	-0.09166
0.79782	-0.11977	-0.09982
0.78339	-0.12901	-0.10711
0.76896	-0.13825	-0.11644
0.75452	-0.14753	-0.13188
0.74008	-0.15602	-0.15704
0.72564	-0.16485	-0.18563
0.7112	-0.17329	-0.19277
0.69675	-0.1814	-0.1962
0.6823	-0.18952	-0.20088
0.66785	-0.19769	-0.20627
0.6534	-0.20522	-0.21202

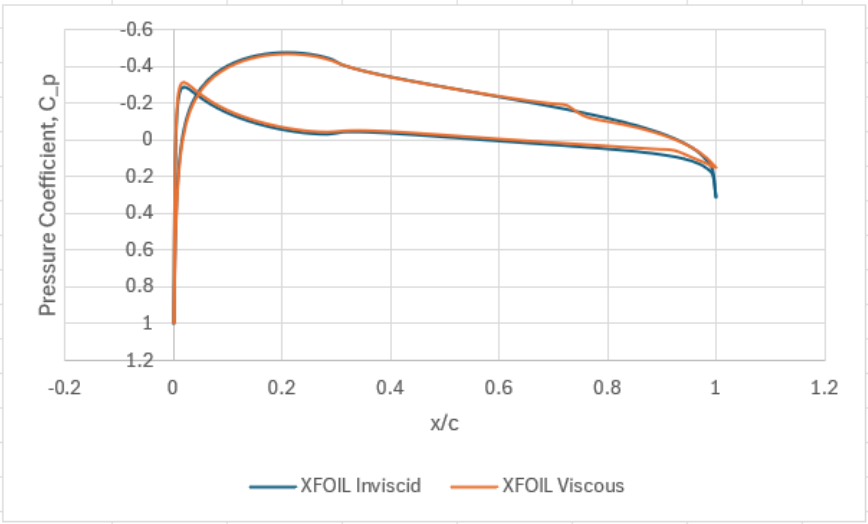


Fig. A16. EXCEL Pressure-Coefficient Distributions for NACA 2308 at α=0°

AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI
x/c	XFOIL Invis	XFOIL Viscous									
1	0.32127	0.13328									
0.99431	0.16739	0.11756									
0.9833	0.10821	0.08851									
0.97014	0.06107	0.0578									
0.9561	0.02528	0.029									
0.94181	-0.00385	0.00307									
0.92746	-0.02909	-0.02047									
0.91308	-0.05147	-0.04197									
0.89869	-0.07219	-0.06194									
0.8843	-0.09028	-0.08018									
0.8699	-0.10785	-0.09759									
0.85549	-0.12536	-0.11445									
0.84108	-0.13991	-0.12967									
0.82667	-0.15607	-0.14514									
0.81225	-0.16956	-0.15922									
0.79782	-0.18481	-0.17377									
0.78339	-0.19763	-0.18706									
0.76896	-0.21154	-0.20059									
0.75452	-0.22456	-0.21363									
0.74008	-0.23722	-0.22628									
0.72564	-0.25036	-0.23916									
0.7112	-0.26294	-0.25161									
0.69675	-0.27543	-0.26401									
0.6823	-0.28755	-0.27608									
0.66785	-0.30077	-0.28872									
0.6534	-0.31251	-0.30052									

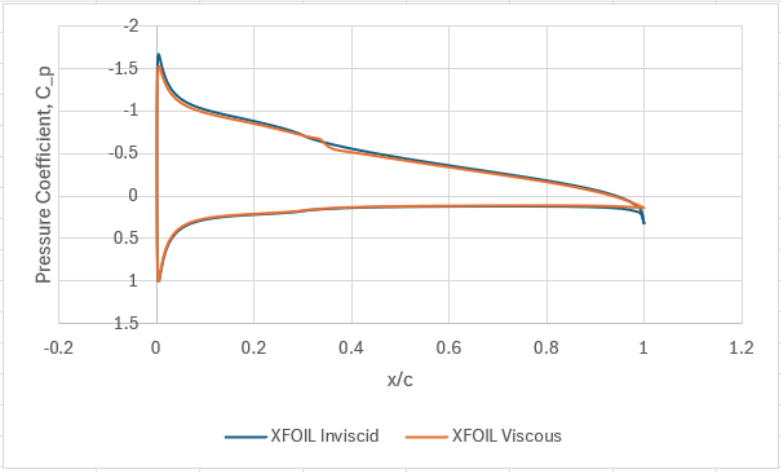


Fig. A17. EXCEL Pressure-Coefficient Distributions for NACA 2308 at α=4°

BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV
x/c	XFOIL Invis	XFOIL Viscous								
1	0.33464	0.10951								
0.99431	0.17427	0.10003								
0.9833	0.11091	0.08156								
0.97014	0.05787	0.05915								
0.9561	0.01663	0.03543								
0.94181	-0.01718	0.01176								
0.92746	-0.0466	-0.01152								
0.91308	-0.07355	-0.0343								
0.89869	-0.09861	-0.0565								
0.8843	-0.12063	-0.07787								
0.8699	-0.14231	-0.0988								
0.85549	-0.16419	-0.11932								
0.84108	-0.18245	-0.13892								
0.82667	-0.20299	-0.15846								
0.81225	-0.22032	-0.17713								
0.79782	-0.23984	-0.19588								
0.78339	-0.2563	-0.21386								
0.76896	-0.27498	-0.23188								
0.75452	-0.2918	-0.24953								
0.74008	-0.30874	-0.26701								
0.72564	-0.32633	-0.28444								
0.7112	-0.34317	-0.3017								
0.69675	-0.36021	-0.31888								
0.6823	-0.37645	-0.33587								
0.66785	-0.39495	-0.35348								
0.6534	-0.41107	-0.37041								

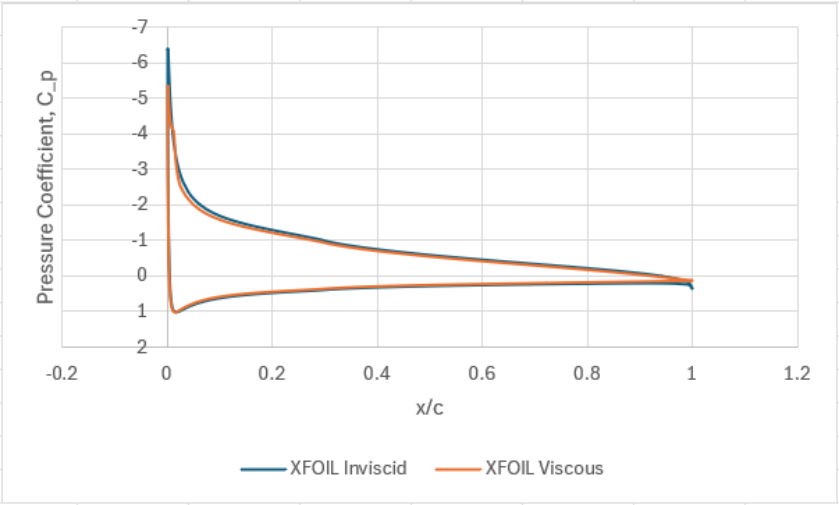


Fig. A18. EXCEL Pressure-Coefficient Distributions for NACA 2308 at α=8°

	AOA	Cl 1108	Cl 4408	Cl 0008	cl 1108 xfoil	1108 Inv cl	4408	4408 inv	0008 inv-xfoil	
-0.0873	-5	-0.4545	-0.0927	-0.5483	-0.4608	-0.4826	-0.0934	-0.094	-0.5847	-0.5567
-0.0785	-4.5	-0.3997	-0.0379	-0.4935	-0.4051	-0.4242	-0.0367	-0.0352	-0.5264	-0.502
-0.0698	-4	-0.3448	0.01694	-0.4386	-0.3494	-0.3657	0.02	0.0235	-0.468	-0.4466
-0.0611	-3.5	-0.29	0.07178	-0.3838	-0.2938	-0.3072	0.0768	0.0822	-0.4096	-0.391
-0.0524	-3	-0.2352	0.12661	-0.329	-0.238	-0.2487	0.1335	0.1409	-0.3511	-0.3352
-0.0436	-2.5	-0.1803	0.18144	-0.2742	-0.1816	-0.1902	0.1902	0.1995	-0.2927	-0.2794
-0.0349	-2	-0.1255	0.23627	-0.2193	-0.1252	-0.1316	0.247	0.2582	-0.2342	-0.2235
-0.0262	-1.5	-0.0707	0.2911	-0.1645		-0.073	0.3038	0.3169	-0.1757	-0.1676
-0.0175	-1	-0.0159	0.34593	-0.1097	-0.0689	-0.0145	0.3606	0.3755	-0.1171	-0.1118
-0.0087	-0.5	0.03898	0.40076	-0.0548	0.0439	0.0441	0.4174	0.4341	-0.0586	-0.0559
0	0	0.09381	0.45559	0	0.1001	0.1027	0.4743	0.4926	-0.0001	0
0.00873	0.5	0.14864	0.51042	0.05483	0.155	0.1612	0.5311	0.5512	0.0585	0.0559
0.01745	1	0.20347	0.56526	0.10966	0.2107	0.2198	0.5878	0.6096	0.117	0.1118
0.02618	1.5	0.2583	0.62009	0.16449	0.2665	0.2783	0.6443	0.6681	0.1755	0.1676
0.03491	2	0.31313	0.67492	0.21932	0.3221	0.3368	0.7002	0.7265	0.2341	0.2235
0.04363	2.5	0.36796	0.72975	0.27416	0.3782	0.3953	0.7542	0.7848	0.2926	0.2793
0.05236	3	0.42279	0.78458	0.32899	0.4345	0.4537	0.8075	0.843	0.351	0.3352
0.06109	3.5	0.47763	0.83941	0.38382	0.4905	0.5122	0.8593	0.9012	0.4095	0.391
0.06981	4	0.53246	0.89424	0.43865	0.5461	0.5705	0.9087	0.9594	0.4679	0.4466
0.07854	4.5	0.58729	0.94907	0.49348	0.6002	0.6289	0.9614	1.0174	0.5263	0.502
0.08727	5	0.64212	1.00391	0.54831		0.6871	1.015	1.0754	0.5846	0.5566
0.09599	5.5	0.69695	1.05874	0.60314	0.6527	0.7454	1.0683	1.1333	0.6429	0.6105
0.10472	6	0.75178	1.11357	0.65797	0.7699	0.8035	1.1218	1.1911	0.7012	0.6631
0.11345	6.5	0.80661	1.1684	0.7128	0.826	0.8617	1.1749	1.2488	0.7594	0.7198
0.12217	7	0.86144	1.22323	0.76764	0.8797	0.9197	1.2278	1.3065	0.8175	0.781
0.1309	7.5	0.91627	1.27806	0.82247	0.9337	0.9777	1.2799	1.364	0.8756	0.8411
0.13963	8	0.97111	1.33289	0.8773	0.9875	1.0356	1.3318	1.4214	0.9336	0.8936
0.14835	8.5	1.02594	1.38772	0.93213	1.041	1.0934	1.3825	1.4787	0.9915	
0.15708	9	1.08077	1.44255	0.98696	1.0945	1.1512	1.4332	1.5359	1.0494	0.9466
0.16581	9.5	1.1356	1.49739	1.04179	1.1476	1.2088	1.4831	1.593	1.1072	1.0531
0.17453	10	1.19043	1.55222	1.09662	1.1998	1.2664	1.5315	1.65	1.1649	1.1062
0.18326	10.5	1.24526	1.60705	1.15145		1.3238	1.5786	1.7068	1.2225	1.1587
0.19199	11	1.30009	1.66188	1.20628	1.3036	1.3812	1.625	1.7635	1.28	1.2104
0.20071	11.5	1.35492	1.71671	1.26112	1.3547	1.4385	1.6711	1.8201	1.3374	1.2609
0.20944	12	1.40976	1.77154	1.31595	1.4045	1.4956	1.7157	1.8765	1.3948	1.3107
0.21817	12.5	1.46459	1.82637	1.37078	1.4532	1.5526	1.7584	1.9328	1.452	1.3594
0.22689	13	1.51942	1.8812	1.42561	1.5008	1.6096	1.7982	1.989	1.5091	1.4065
0.23562	13.5	1.57425	1.93603	1.48044		1.6664	1.8342	2.045	1.5661	1.4546
0.24435	14	1.62908	1.99087	1.53527	1.5943	1.723	1.8635	2.1008	1.623	1.502
0.25307	14.5	1.68391	2.0457	1.5901	1.6394	1.7796	1.8851	2.1565	1.6797	1.548
0.2618	15	1.73874	2.10053	1.64493	1.683	1.836	1.9036	2.212	1.7364	1.5922
	15.5								1.7929	1.6349
	16								1.8492	1.6752
	16.5								1.9055	1.7129
	17								1.9615	1.7469
	17.5								2.0175	1.7752
	18								2.0733	1.7873
	18.5								2.1289	1.7905
	19								2.1843	1.7872
	19.5								2.2396	1.773
	20								2.2948	1.7404

Fig. A19. EXCEL NACA Airfoil Lift-Curve Comparison

VII. References

1. Anderson, J. D., *Fundamentals of Aerodynamics*, 7th ed., McGraw Hill, 2024.
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